

Additional Tables and Diagnostics for “What a Pooled Leaderboard Does Not Say: Auditing the Scope and Provenance of Method Comparisons in Microservice RCA Benchmarks”

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This document gives additional definitions, run details, diagnostics, and complete result tables for the paper. Section numbers of the paper are written as “paper §”. All tables in Section 12 are generated from the analysis outputs of the frozen per-case table; the diagnostic tables in Sections 5 to 7 are generated from the stored outputs of the diagnostic runs named in their captions. Accuracy is always computed over all cases of a subsystem, with invalid outputs scored 0, unless a column says otherwise.

1 Probe Definitions

All three probes receive the same input as the published methods on each family: a table of metric time series, the injection (or query) time t_0 , and a mapping from each metric to its component (the service prefix of the metric name on RCAEval, the node on PetShop, and the component parser of the OpenRCA wrapper in Section 2). Let $x_m(t)$ be metric m , B the time points before t_0 in the analysis window, and A the time points at or after t_0 . Non-numeric columns and the time column are ignored.

max-|Z|. For each metric, μ_m and σ_m are the mean and the population standard deviation of x_m over B , and the score is $z_m = \max_{t \in A} |x_m(t) - \mu_m| / \sigma_m$. Metrics with $\sigma_m = 0$, or with no finite value in A , score 0. Metrics are ranked by z_m in descending order, ties by name; the answer is the component of the first metric (on OpenRCA, of the first metric that belongs to a candidate component; Section 2).

alert-count. With z_m as above, a metric raises an alert if $z_m > 3$. Components are ranked by their number of alerting metrics in descending order, ties by name; within a component, metrics are ordered by z_m . The answer is the first component.

CD-1min. For each metric with at least 10 finite baseline values, μ_m and s_m are the mean and sample standard deviation over B (metrics with $s_m < 10^{-8}$ are skipped). The peak p_m is the value in A farthest from μ_m (the maximum or the minimum). The metric is flagged if both $|p_m - \mu_m| / s_m \geq 3$ and $100 |p_m - \mu_m| / (|\mu_m| + 10^{-8}) \geq 5$. Each component receives the largest relative change among its flagged metrics, and components are ranked by it (ties by the largest z , then by name). If no metric is flagged, the output is empty and counts as invalid. On OpenRCA, CD-1min runs on one-minute data with a baseline of 4 hours before a detection window that starts 30 minutes before the query midpoint and ends one minute after it; this is the resolution for which it was written.

2 OpenRCA Wrapper and Invalid Outputs

OpenRCA asks for a root-cause component, a reason, and an occurrence time, while the audited methods rank metrics. For each query, the wrapper loads the metrics that cover the query window (a 4-hour baseline and 30 minutes on either side of the window midpoint) and passes them to the method. The answer component is the component of the first ranked metric that belongs to the system’s candidate component list; the reason is mapped from the type of that metric; the time is the moment at which that metric deviates most from its baseline. The official scorer then awards the fraction of the required items that match (component and reason by exact string, time within 60 s).

When the wrapper cannot produce an answer (no data for the window, an exception in the method, an empty ranking, or no candidate component in the ranking), it returns a fixed answer per system. Such fixed answers can earn partial credit, because every query is scored on each required item separately. We therefore record the wrapper status of every query and treat every fixed answer as an invalid output that scores 0 (paper §3.3). This matters most for ϵ -Diagnosis on Telecom: the method returns exactly three metrics per query, none of which belonged to a candidate component in 32 of the 51 queries. The per-subsystem coverage of every method is listed in Table 12. One earlier run of alert-count on OpenRCA passed a wrongly named keyword argument, so every query fell back to the fixed answer; the main analysis uses the corrected run, and the earlier run is kept only as a diagnostic in the released table.

3 Under-Specified Choices and Silent Failures in RCAEval

This section gives the details of the under-specified input choice and the two silent failures that paper §3.3 and §5.2 discuss; all refer to the audited repository version (commit 497505b).

Input file. For RE1-SS and RE1-TT, each case directory contains a raw export (`data.csv`) and a reduced table (`simple_data.csv`). The released runner globs for `data.csv` first and falls back to `simple_metrics.csv` (the RE2 file name); the lines that would switch to `simple_data.csv` are commented out. RE1-OB has only `data.csv`, with 51–60 columns. Accuracy of every method on both inputs is in Table 13.

Dependency version. CausalRCA ranks the nodes of its learned graph with the PageRank implementation of scikit-network. The `fit_transform` method it calls was removed in scikit-network 0.32 (as stated in the maintainers’ later fix); the released requirements set only a lower bound. Under a newer version, every call raises an exception that the wrapper catches, and the wrapper returns the input column order. In a permutation test on 30 cases (10 each from RE1-OB, RE2-OB, and RE2-SS), the ranking equalled the input column order in 30 of 30 cases under scikit-network 0.33.0, and after permuting the columns it equalled the permuted order in 30 of 30; under 0.31.0 it equalled the column order in 0 of 30. All CausalRCA results in the paper use 0.31.0.

Duplicated header. Fifty of the 125 RE1-OB tables (all 25 CPU and all 25 memory faults) contain a second column named `time`, which pandas reads as `time.1`. No other RCAEval input file we use has a repeated column name. On the audited release, CIRCA returned the input column order in 49 of the 50 affected cases and in none of the 75 unaffected cases; ϵ -Diagnosis ranked `time.1` first in 49 of the 50. We reran CIRCA, ϵ -Diagnosis, and RCD (three seeds) on all 125 RE1-OB cases with the duplicated column dropped when the table is read, which is the same change as the upstream fix. When CIRCA was rerun on all 125 RE1-OB cases, 1 of the 75 unaffected cases gave a different top-1 than the older run, which came from a different run script and environment; we kept the older values for those 75 cases and replaced only the 50 affected ones, so the fixed condition combines two runs that differ in one case. On the 50 affected cases, CIRCA’s top-1 accuracy rose from 0.220 to 0.820 (32 case scores changed); ϵ -Diagnosis’s scores did not change (0.000 before and after); RCD changed 5, 4, and 3 case scores for its three seeds. BARO and the three probes gave identical scores with and without the fix, so their original runs are kept. The as-shipped outputs remain in the released table under a separate role and are compared with the fixed outputs in Table 15.

4 CausalRCA Runs

Seeds. The RCAEval version of CausalRCA defines a seed but never sets it, and in the original implementation the seed calls are commented out; the original repository’s scripts repeat every configuration ten times. We set the Python, NumPy, and PyTorch seeds to 0, 1, and 2. Rerunning 18 case–seed combinations reproduced the top-20 ranking of the main run in all 18, so the runs are reproducible given the seed, and the differences between seeds come from the random initialization. Per-seed accuracy varies widely, for example 0.160, 0.208, and 0.040 on RE1-TT; per-seed pair results are in Table 9.

Empty graphs. The learned adjacency matrix is thresholded at 0.3. On RE2-TT with seed 1, 43 of the 90 thresholded graphs were empty; PageRank then raises an error and the wrapper returns the column order. We count these 43 outputs as invalid (coverage in Table 12). No other dataset or seed returned the column order.

Difference from the original implementation. The RCAEval version divides every metric by its column maximum before training; the original training scripts do not normalize. Hyperparameters, network architecture, and the thresholding steps are identical. On RE2-TT our three-seed mean top-1 accuracy is 0.207, against 0.22 in RCAEval’s Table 6.

5 Authors’ SimpleRCA Branch

The default branch of the authors’ repository is not the code behind the RCAEval results in the SimpleRCA paper. Two other branches are: one for RE2 and RE3 (commit 5fd1f7f), and one that changes the preprocessing and BARO setting for RE3-SS (commit 7f7f295). Both run on a benchmark platform whose data converter must produce log tables with an error attribute that the code reads; the version pinned in the authors’ lock file does not, and the first platform release that does was published about seven hours after the branch commit. We copy the algorithm files unchanged, replace only the platform input and output layer, and convert the data with that platform release. Table 1 shows that this reproduces every value the authors report for SimpleRCA and their BARO baseline on RE2 and RE3 to two decimals.

The branch ranks services by BARO’s metric scores (the maximum per service), and then adds points from trace and log evidence to the top three services. This differs from the procedure described in the paper’s text, which counts threshold alerts per service. RE1 contains only metrics, so on RE1 the branch adds no trace or log evidence; its RE1 results (bottom of Table 1) are therefore reported here and not as a separate method in the main analysis. They are close to those of the authors’ own BARO adapter on RE1-OB and RE1-SS and higher on RE1-TT. The RE1-OB inputs of these runs include the 50 tables with the duplicated header; we did not rerun them, because they enter no comparison.

The branch’s code differs from the method description in the SimpleRCA paper’s Section 3.1.2 in details of the scoring step; we use the code as released because it reproduces the authors’ reported values, and we do not attempt to reconcile the text.

Table 1: Authors’ SimpleRCA branch. Top: our runs on RE2 and RE3 with the values reported in the SimpleRCA paper in parentheses (Top@1, Top@3, Top@5, Avg@3, Avg@5, MRR with misses counted as 0); the last column is the largest absolute difference after rounding to two decimals. RE3-SS uses the second branch the authors provide. Bottom: the same code on RE1 (20-minute window; OB data.csv, SS/TT simple_data.csv), scored by exact service-name match; RE1 contains only metrics, so no trace or log evidence is added. Source: `simplerca_summary.txt`.

Data	Method	Top@1	Top@3	Top@5	Avg@3	Avg@5	MRR	Max diff.
RE2-OB	BARO (authors’)	0.14 (0.14)	0.93 (0.93)	0.98 (0.98)	0.64 (0.64)	0.77 (0.77)	0.54 (0.54)	0.00
RE2-OB	SimpleRCA	0.47 (0.47)	0.93 (0.93)	1.00 (1.00)	0.68 (0.68)	0.80 (0.80)	0.67 (0.67)	0.00
RE2-SS	BARO (authors’)	0.14 (0.14)	0.82 (0.82)	0.94 (0.94)	0.50 (0.50)	0.67 (0.67)	0.47 (0.47)	0.00
RE2-SS	SimpleRCA	0.24 (0.24)	0.88 (0.88)	0.97 (0.97)	0.59 (0.59)	0.74 (0.74)	0.55 (0.55)	0.00
RE2-TT	BARO (authors’)	0.67 (0.67)	0.84 (0.84)	0.89 (0.89)	0.77 (0.77)	0.81 (0.81)	0.76 (0.76)	0.00
RE2-TT	SimpleRCA	0.80 (0.80)	0.87 (0.87)	0.93 (0.93)	0.84 (0.84)	0.87 (0.87)	0.85 (0.85)	0.00
RE3-OB	BARO (authors’)	0.00 (0.00)	0.90 (0.90)	0.90 (0.90)	0.59 (0.59)	0.71 (0.71)	0.45 (0.45)	0.00
RE3-OB	SimpleRCA	0.30 (0.30)	0.90 (0.90)	0.90 (0.90)	0.62 (0.62)	0.73 (0.73)	0.56 (0.56)	0.00
RE3-SS	BARO (authors’)	0.00 (0.00)	0.83 (0.83)	0.93 (0.93)	0.46 (0.46)	0.65 (0.65)	0.40 (0.40)	0.00
RE3-SS	SimpleRCA	0.83 (0.83)	0.90 (0.90)	0.90 (0.90)	0.88 (0.88)	0.89 (0.89)	0.87 (0.87)	0.00
RE3-TT	BARO (authors’)	0.50 (0.50)	0.93 (0.93)	1.00 (1.00)	0.77 (0.77)	0.86 (0.86)	0.72 (0.72)	0.00
RE3-TT	SimpleRCA	0.83 (0.83)	1.00 (1.00)	1.00 (1.00)	0.93 (0.93)	0.96 (0.96)	0.91 (0.91)	0.00
RE1-OB	BARO (authors’)	0.776	0.928	0.968	0.867	0.904	0.860	–
RE1-OB	SimpleRCA	0.784	0.936	0.976	0.875	0.912	0.864	–
RE1-SS	BARO (authors’)	0.368	0.920	0.992	0.680	0.800	0.634	–
RE1-SS	SimpleRCA	0.368	0.928	0.992	0.683	0.803	0.634	–
RE1-TT	BARO (authors’)	0.464	0.672	0.728	0.581	0.634	0.581	–
RE1-TT	SimpleRCA	0.568	0.856	0.952	0.733	0.813	0.721	–

6 Comparison with RCAEval’s Published RE2 Table

RCAEval’s paper reports per-fault accuracy only for RE2-TT (its Table 6). Table 2 lists our runs next to it. For TraceRCA and MicroRank, the 18 per-fault cells computed over valid outputs only are within 0.01 of Table 6, while the same accuracy over all cases is lower by up to 0.41 on the memory and socket faults: both methods raise an error on the same 17 of the 90 cases, because a service–operation pair that appears only after the injection has no baseline entry. The multi-source variant of BARO uses its trace input only if the dataset name passed to it is `mm-tt`; with the name that the released runner passes (`re2-tt`), its AC@1 is almost unchanged (0.68 against 0.69), and with the raw metric export instead of the reduced one it drops to 0.14. CIRCA, run with the configuration shipped in the repository, has a higher AC@1 than Table 6 on five of the six faults and the same on the sixth; we could not identify the configuration behind the published values.

Table 2: RE2-TT per fault type: our runs over all cases (invalid outputs score 0), over valid outputs only, and the values in RCAEval’s Table 6. Cells are AC@1 / AC@3 / Avg@5. Source: `re2_table6_compare.txt`.

Fault	All cases	Valid outputs only	Table 6
CIRCA: 90/90 valid outputs			
cpu	0.40 / 0.60 / 0.53 ($n=15$)	0.40 / 0.60 / 0.53 ($n=15$)	0.27 / 0.27 / 0.28
mem	0.73 / 0.93 / 0.88 ($n=15$)	0.73 / 0.93 / 0.88 ($n=15$)	0.47 / 0.73 / 0.68
disk	0.80 / 0.80 / 0.80 ($n=15$)	0.80 / 0.80 / 0.80 ($n=15$)	0.53 / 0.67 / 0.64
socket	0.80 / 0.87 / 0.84 ($n=15$)	0.80 / 0.87 / 0.84 ($n=15$)	0.27 / 0.53 / 0.52
delay	0.20 / 0.40 / 0.36 ($n=15$)	0.20 / 0.40 / 0.36 ($n=15$)	0.2 / 0.27 / 0.28
loss	0.33 / 0.60 / 0.56 ($n=15$)	0.33 / 0.60 / 0.56 ($n=15$)	0.2 / 0.33 / 0.35
all	0.54 / 0.70 / 0.66 ($n=90$)	0.54 / 0.70 / 0.66 ($n=90$)	0.32 / 0.47 / 0.46
MicroRank: 73/90 valid outputs			
cpu	0.20 / 0.40 / 0.32 ($n=15$)	0.21 / 0.43 / 0.34 ($n=14$)	0.21 / 0.43 / 0.34
mem	0.13 / 0.20 / 0.17 ($n=15$)	0.25 / 0.38 / 0.33 ($n=8$)	0.25 / 0.38 / 0.33
disk	0.00 / 0.33 / 0.25 ($n=15$)	0.00 / 0.36 / 0.27 ($n=14$)	0 / 0.36 / 0.27

Fault	All cases	Valid outputs only	Table 6
socket	0.20 / 0.27 / 0.24 ($n=15$)	0.30 / 0.40 / 0.36 ($n=10$)	0.3 / 0.4 / 0.36
delay	0.07 / 0.27 / 0.20 ($n=15$)	0.08 / 0.31 / 0.23 ($n=13$)	0.08 / 0.31 / 0.23
loss	0.13 / 0.33 / 0.28 ($n=15$)	0.14 / 0.36 / 0.30 ($n=14$)	0.14 / 0.36 / 0.3
all	0.12 / 0.30 / 0.24 ($n=90$)	0.15 / 0.37 / 0.30 ($n=73$)	0.16 / 0.37 / 0.31
BARO (multi-source), name mm-tt: 89/90 valid outputs			
cpu	0.47 / 0.80 / 0.75 ($n=15$)	0.47 / 0.80 / 0.75 ($n=15$)	0.47 / 0.8 / 0.75
mem	0.93 / 1.00 / 0.99 ($n=15$)	0.93 / 1.00 / 0.99 ($n=15$)	0.93 / 1 / 0.99
disk	1.00 / 1.00 / 1.00 ($n=15$)	1.00 / 1.00 / 1.00 ($n=15$)	1 / 1 / 1
socket	0.60 / 0.73 / 0.73 ($n=15$)	0.64 / 0.79 / 0.79 ($n=14$)	0.6 / 0.8 / 0.79
delay	0.47 / 0.67 / 0.61 ($n=15$)	0.47 / 0.67 / 0.61 ($n=15$)	0.47 / 0.67 / 0.61
loss	0.67 / 0.67 / 0.71 ($n=15$)	0.67 / 0.67 / 0.71 ($n=15$)	0.67 / 0.67 / 0.71
all	0.69 / 0.81 / 0.80 ($n=90$)	0.70 / 0.82 / 0.81 ($n=89$)	0.69 / 0.82 / 0.81
BARO (multi-source), metrics.csv: 89/90 valid outputs			
cpu	0.00 / 0.00 / 0.00 ($n=15$)	0.00 / 0.00 / 0.00 ($n=15$)	0.47 / 0.8 / 0.75
mem	0.00 / 0.07 / 0.16 ($n=15$)	0.00 / 0.07 / 0.16 ($n=15$)	0.93 / 1 / 0.99
disk	0.80 / 1.00 / 0.96 ($n=15$)	0.80 / 1.00 / 0.96 ($n=15$)	1 / 1 / 1
socket	0.00 / 0.00 / 0.00 ($n=15$)	0.00 / 0.00 / 0.00 ($n=14$)	0.6 / 0.8 / 0.79
delay	0.07 / 0.13 / 0.17 ($n=15$)	0.07 / 0.13 / 0.17 ($n=15$)	0.47 / 0.67 / 0.61
loss	0.00 / 0.07 / 0.04 ($n=15$)	0.00 / 0.07 / 0.04 ($n=15$)	0.67 / 0.67 / 0.71
all	0.14 / 0.21 / 0.22 ($n=90$)	0.15 / 0.21 / 0.22 ($n=89$)	0.69 / 0.82 / 0.81
BARO (multi-source), name re2-tt: 89/90 valid outputs			
cpu	0.47 / 0.80 / 0.75 ($n=15$)	0.47 / 0.80 / 0.75 ($n=15$)	0.47 / 0.8 / 0.75
mem	0.93 / 1.00 / 0.99 ($n=15$)	0.93 / 1.00 / 0.99 ($n=15$)	0.93 / 1 / 0.99
disk	1.00 / 1.00 / 1.00 ($n=15$)	1.00 / 1.00 / 1.00 ($n=15$)	1 / 1 / 1
socket	0.53 / 0.73 / 0.72 ($n=15$)	0.57 / 0.79 / 0.77 ($n=14$)	0.6 / 0.8 / 0.79
delay	0.47 / 0.67 / 0.61 ($n=15$)	0.47 / 0.67 / 0.61 ($n=15$)	0.47 / 0.67 / 0.61
loss	0.67 / 0.67 / 0.71 ($n=15$)	0.67 / 0.67 / 0.71 ($n=15$)	0.67 / 0.67 / 0.71
all	0.68 / 0.81 / 0.80 ($n=90$)	0.69 / 0.82 / 0.80 ($n=89$)	0.69 / 0.82 / 0.81
TraceRCA: 73/90 valid outputs			
cpu	0.60 / 0.73 / 0.69 ($n=15$)	0.64 / 0.79 / 0.74 ($n=14$)	0.64 / 0.79 / 0.74
mem	0.33 / 0.47 / 0.44 ($n=15$)	0.62 / 0.88 / 0.82 ($n=8$)	0.63 / 0.88 / 0.83
disk	0.60 / 0.67 / 0.69 ($n=15$)	0.64 / 0.71 / 0.74 ($n=14$)	0.64 / 0.71 / 0.74
socket	0.40 / 0.53 / 0.51 ($n=15$)	0.60 / 0.80 / 0.76 ($n=10$)	0.6 / 0.8 / 0.76
delay	0.73 / 0.73 / 0.76 ($n=15$)	0.85 / 0.85 / 0.88 ($n=13$)	0.85 / 0.85 / 0.88
loss	0.53 / 0.67 / 0.63 ($n=15$)	0.57 / 0.71 / 0.67 ($n=14$)	0.57 / 0.71 / 0.67
all	0.53 / 0.63 / 0.62 ($n=90$)	0.66 / 0.78 / 0.76 ($n=73$)	0.66 / 0.79 / 0.77

7 Window Length on RE1

All RE1 and RE2 runs in the paper use RCAEval’s default window of 20 minutes (10 minutes before and after the injection). Table 3 compares it with a 10-minute window for BARO and the three probes. The accuracy changes by up to 0.208 (max- $|Z|$ on RE1-OB); we report this as a diagnostic and do not use the 10-minute runs in any comparison.

8 Margin and Spread

The paper describes comparisons by margin and spread only to read the forest figure; this section gives the counts. A pooled comparison can be described by its margin $|\Delta|$ and the spread of its subsystem differences, and the audited comparisons sort along this description. Among published pairs, 8 have a margin smaller than the spread; 6 of them reverse, 2 with CI support, whereas of the 15 whose margin exceeds the spread 1 reverses, without CI support (with probes and shipped baselines: 25 of 34, and 2 of 64). The same split appears in two tables that the benchmarks’ authors published: in RCAEval’s RE2-TT accuracy by fault type (RCAEval paper, Table 6), 21 of the 23 pairs with margin below spread reverse on some fault type, against 2 of 32 above it; in PetShop’s accuracy by scenario and fault type (PetShop paper, Table 8), 6 of 8 and 0 of 7. The split is partly definitional,

Table 3: RE1 top-1 accuracy with a 10- and a 20-minute window (125 cases each). Lost / gained: cases correct at 10 minutes and wrong at 20, and the reverse. Source: `re1_window_compare.txt`.

Method	Subsystem	10 min	20 min	Lost	Gained	Valid (10/20)
BARO	RE1-OB	0.728	0.736	3	4	125/125
BARO	RE1-SS	0.832	0.848	1	3	125/125
BARO	RE1-TT	0.592	0.560	9	5	125/125
max- Z	RE1-OB	0.712	0.504	32	6	125/125
max- Z	RE1-SS	0.840	0.744	16	4	125/125
max- Z	RE1-TT	0.448	0.424	9	6	125/125
alert-count	RE1-OB	0.432	0.392	7	2	125/125
alert-count	RE1-SS	0.320	0.376	3	10	125/125
alert-count	RE1-TT	0.072	0.048	5	2	125/125
CD-1min	RE1-OB	0.792	0.728	8	0	125/125
CD-1min	RE1-SS	0.944	0.928	3	1	125/125
CD-1min	RE1-TT	0.304	0.280	4	1	125/125

since a pair whose differences change sign has a spread comparable to its mean, and it describes these samples rather than giving a selection rule: applied to the remaining subsystems in the held-out analysis, it did not reduce the 8 wrong choices (Section 9).

9 Held-Out Subsystems

For each family, each subsystem is held out in turn; methods present on every subsystem of the family are ranked by their mean over the remaining subsystems (subsystems weighted equally, ties by name), and every adjacent pair of that ranking is one decision. Margin and spread are computed from the remaining subsystems only, so the spread rests on two differences for RE1 and three for OpenRCA and PetShop. Table 4 gives the counts for several multiples c of the spread, by layer and family.

10 Cross-Family Comparison Set

Six methods have outputs on all 11 subsystems of the main analysis: BARO, RCD, ϵ -Diagnosis, and the three probes (on PetShop, RCD and ϵ -Diagnosis are the implementations PetShop ships). Table 5 lists, for each of the 15 pairs, the held-out subsystems on which the pooled winner of the other ten is less accurate; Table 6 gives the robustness checks summarized in the main text.

11 Survey of Reporting Practice

Selection. We listed 26 microservice RCA method and benchmark papers from 2021 to 2026: the papers of the audited benchmarks and methods, and papers found by keyword search in software-engineering, systems, and machine-learning venues, with at most five papers per research group. This is a convenience sample, not a systematic review. Full text was obtained for 23 papers; the other three (MicroRank, Nezha, and MRCA) are listed in Table 20 but excluded from every count in the paper and in this document.

Per-system reporting. For each of the 23 papers we recorded whether it evaluates more than one system and, if so, whether it reports a score for each system. 17 papers evaluate more than one system or dataset (simulation datasets of different sizes count as datasets), and all 17 report a score per system; 6 evaluate one system, among them PetShop, whose three traffic scenarios are one application. Table 20 gives the result and the table, figure, or passage checked for each paper.

Release check. On 2 and 3 October 2026 we listed the files of every repository and archive linked from each paper, at the current commit, and recorded whether the release contains per-case predictions or scores of any method, which methods and cases they cover, and whether they suffice to recompute every pairwise comparison of the paper’s main results. 7 papers release per-case outputs of at least one method. For 3 (BARO, RCAEval, and the study howfar2024, which shares RCAEval’s repository) the only such outputs are the ones a tutorial notebook saved for a single case, and they appear in the current repository, not necessarily in the one at publication. 3 release the outputs of the authors’ own method on their test cases: OpenRCA’s repository holds one archived run of its agent on all 335 queries, whose files record neither the model nor the setting; CausalRCA’s repository holds the rankings of the authors’ method as text files per fault type and setting; and the archive behind CHASE’s link, which we downloaded, holds the predictions of the authors’ method on the 939 test cases of its split. CIRCA’s README links an archive with the rankings of all eleven methods of its simulation study on every case of every run, from which its Table 1 can be recomputed, but only summary files for its production dataset. None of these releases includes outputs of the other methods of the paper’s main comparison on every dataset. For 2 papers a linked release had expired or could not be reached (TraceRCA’s data archive and the anonymous review links of one paper). For six papers the full text gives no link to a release of the authors’

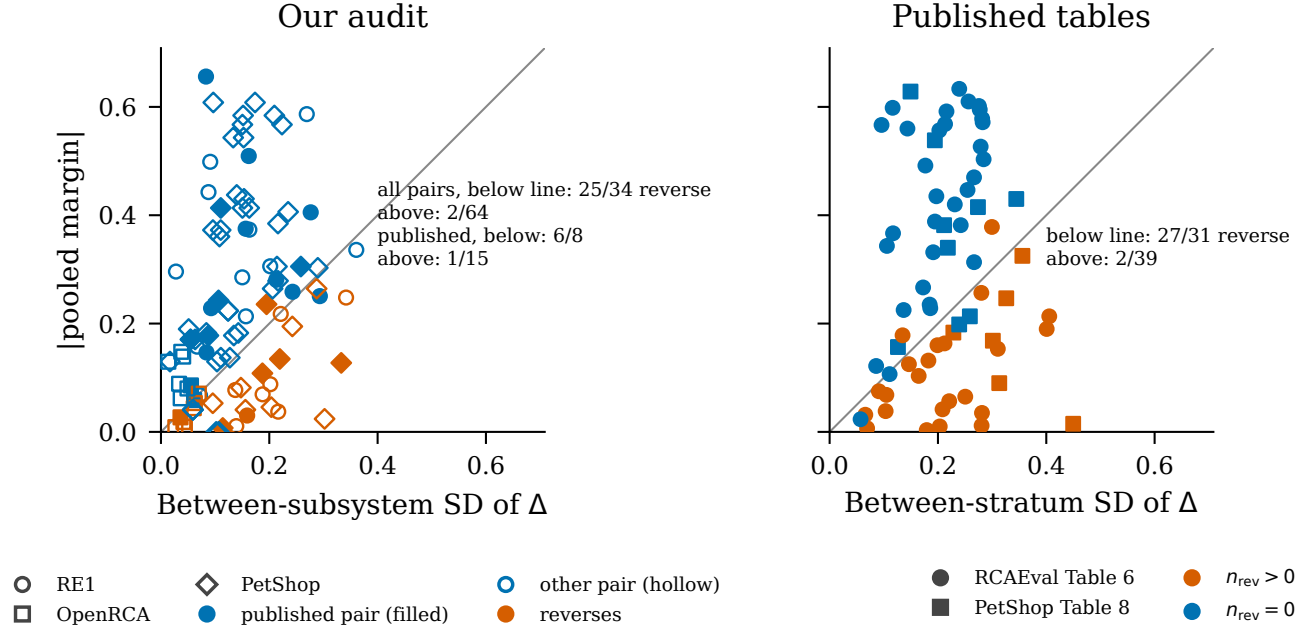


Figure 1: Pooled margin $|\bar{\Delta}|$ against between-subsystem spread (sample standard deviation of Δ_s). Left: every pair in the main analysis, coloured by family (filled: pairs of published methods); orange marks pairs that reverse on some subsystem. Right: pairs inside two published tables, RCAEval Table 6 (RE2-TT by fault type) and PetShop Table 8 (by scenario and fault type). The diagonal is margin = spread.

own, which we record as no release linked rather than as no outputs; for the remaining papers we examined the linked repositories and, where linked, the data archives, without an exhaustive scan of nested archives. Table 7 lists the result per paper; the notes in the package give every link and the files found.

LLM use. LLM sessions located the tables and passages behind the per-system check and listed the repository contents behind the release check; every statement above was verified by an author against the paper or the repository (paper §3.5).

Table 7: Released per-case outputs in the 23 surveyed papers with full text, checked on 2 and 3 October 2026. Any: the release contains per-case predictions or scores of at least one method; What: which outputs. All: it contains them for every method on every dataset of the paper’s main comparison (partial: for every method of one of its experiments). undet.: a linked release had expired or could not be reached. Release: first repository checked, at the commit checked (–: no release linked); the notes in the package give every link and the files found.

Paper	Any	What	All	Release
baro2024	yes	one demonstration case	no	phamquilluan/baro@e35f4ec
circa2022	yes	all methods, simulation study only	partial	NetManAI0ps/CIRCA@0215e18
rcd2022	no	–	no	azamikram/rcd@373882c
tracerca2021	undet.	–	undet.	NetManAI0ps/TraceRCA@b9087c7
rcaeal2025	yes	one demonstration case	no	phamquilluan/RCAEval@259ea41
fang2026	no	–	no	LGU-SE-Internal/Simple-RCA@1be3911
openrca2025	yes	own method, all test cases	no	microsoft/OpenRCA@c1bd4af
petshop2024	no	–	no	amazon-science/petshop-root-cause-analysis@2e96f93
eadro2023	no	–	no	BEbillionaireUSD/Eadro@82ff9c9

Table 7: Released per-case outputs in the 23 surveyed papers with full text, checked on 2 and 3 October 2026. Any: the release contains per-case predictions or scores of at least one method; What: which outputs. All: it contains them for every method on every dataset of the paper’s main comparison (partial: for every method of one of its experiments). undet.: a linked release had expired or could not be reached. Release: first repository checked, at the commit checked (–: no release linked); the notes in the package give every link and the files found.

Paper	Any	What	All	Release
mabc2024	no	–	no	zwpride/mABC@79272ce
mulan2024	no	–	no	–
ocean2024	no	–	no	–
tvdiag2024	no	–	no	WHU-AISE/TVDiag@cf6f334
chase2024	yes	own method, all test cases	no	Drive archive
howfar2024	yes	one demonstration case	no	phamquilian/RCAEval@259ea41
lemmarca2024	no	–	no	zhengzhangchen/LEMMA_rca_baselines@9b56afb
sparserca2024	no	–	no	–
tracecontrast2024	no	–	no	–
tracediag2023	no	–	no	–
run2024	no	–	no	zmlin1998/RUN@e2c97d7
dynacausal2025	no	–	no	–
causalrca2023	yes	own method, all test cases	no	AXinx/CausalRCA_code@21f4eb0
toomanycooks2025	undet.	–	undet.	–

12 Complete Result Tables

All pairs, main analysis

Every seed-averaged method pair, with both pooled differences, reversal counts, dispersion, and the Holm-adjusted likelihood-ratio test. Source: `paper/data/rq1/pairs.csv`.

Table 8: All method pairs in the main analysis (seed-averaged). Pooled Δ (equal) weights subsystems equally and is shown with its 95% CI; the case-weighted pooled Δ weights subsystems by case count. #rev is the number of subsystems whose Δ has the opposite sign of the equal-weight pooled Δ ; #rev CI counts those whose CI excludes 0. Below line marks $|\text{pooled } \Delta| < \text{SD of subsystem } \Delta$. The 95% prediction interval is reported only where the random-effects fit supplies one. LRT p is Holm-adjusted. Sep. marks complete separation.

Pair	k	Pooled Δ (equal) [CI]	Pooled Δ (case-wtd.)	#rev	SD of Δ_s	Below line	95% PI	LRT p (Holm)	Sep.
<i>RCAEval RE1, Published</i>									
BARO vs CausalRCA	3	+0.405 [+0.357, +0.453]	+0.405	0	0	0.276	–	9.53×10^{-9}	
BARO vs CIRCA	3	+0.147 [+0.096, +0.200]	+0.147	0	0	0.084	–	0.204	
BARO vs ϵ -Diagnosis	3	+0.656 [+0.608, +0.704]	+0.656	0	0	0.083	–	0.054	✓
BARO vs RCD	3	+0.375 [+0.319, +0.430]	+0.375	0	0	0.156	–	0.001	
CausalRCA vs CIRCA	3	–0.259 [–0.308, –0.207]	–0.259	0	0	0.243	–	2.30×10^{-6}	
CausalRCA vs ϵ -Diagnosis	3	+0.251 [+0.211, +0.291]	+0.251	0	0	0.293	✓	8.12×10^{-10}	✓
CausalRCA vs RCD	3	–0.030 [–0.077, +0.017]	–0.030	2	0	0.159	✓	0.001	
CIRCA vs ϵ -Diagnosis	3	+0.509 [+0.461, +0.560]	+0.509	0	0	0.162	–	0.017	✓
CIRCA vs RCD	3	+0.228 [+0.167, +0.286]	+0.228	0	0	0.093	–	0.057	
ϵ -Diagnosis vs RCD	3	–0.281 [–0.329, –0.234]	–0.281	0	0	0.213	–	0.001	✓
<i>RCAEval RE1, All</i>									
Alert-count vs BARO	3	–0.443 [–0.501, –0.381]	–0.443	0	0	0.088	–	0.019	
Alert-count vs CausalRCA	3	–0.037 [–0.088, +0.012]	–0.037	1	1	0.216	✓	6.98×10^{-6}	
Alert-count vs CD-1min	3	–0.373 [–0.432, –0.315]	–0.373	0	0	0.163	–	0.019	
Alert-count vs CIRCA	3	–0.296 [–0.357, –0.232]	–0.296	0	0	0.028	–	0.406	
Alert-count vs ϵ -Diagnosis	3	+0.213 [+0.165, +0.261]	+0.213	0	0	0.157	–	0.053	✓
Alert-count vs max- Z	3	–0.285 [–0.347, –0.221]	–0.285	0	0	0.150	–	1.40×10^{-4}	

Continued on next page

Table 8: All method pairs in the main analysis (seed-averaged). Pooled Δ (equal) weights subsystems equally and is shown with its 95% CI; the case-weighted pooled Δ weights subsystems by case count. #rev is the number of subsystems whose Δ has the opposite sign of the equal-weight pooled Δ ; #rev CI counts those whose CI excludes 0. Below line marks $|\text{pooled } \Delta| < \text{SD of subsystem } \Delta$. The 95% prediction interval is reported only where the random-effects fit supplies one. LRT p is Holm-adjusted. Sep. marks complete separation.

Pair	k	Pooled Δ (equal) [CI]	Pooled Δ (case-wtd.)	#rev	#rev CI	SD of Δ_s	Below line	95% PI	LRT p (Holm)	Sep.
Alert-count vs RCD	3	-0.068 [-0.123, -0.012]	-0.068	0	0	0.072	✓	-	0.826	
BARO vs CausalRCA	3	+0.405 [+0.357, +0.453]	+0.405	0	0	0.276		-	2.65×10^{-8}	
BARO vs CD-1min	3	+0.069 [+0.016, +0.120]	+0.069	1	1	0.188	✓	-	0.001	
BARO vs CIRCA	3	+0.147 [+0.096, +0.200]	+0.147	0	0	0.084		-	0.818	
BARO vs ϵ -Diagnosis	3	+0.656 [+0.608, +0.704]	+0.656	0	0	0.083		-	0.131	✓
BARO vs max- Z	3	+0.157 [+0.107, +0.208]	+0.157	0	0	0.067		-	0.880	
BARO vs RCD	3	+0.375 [+0.319, +0.430]	+0.375	0	0	0.156		-	0.002	
CausalRCA vs CD-1min	3	-0.336 [-0.383, -0.290]	-0.336	0	0	0.361	✓	-	1.87×10^{-14}	
CausalRCA vs CIRCA	3	-0.259 [-0.308, -0.207]	-0.259	0	0	0.243		-	6.60×10^{-6}	
CausalRCA vs ϵ -Diagnosis	3	+0.251 [+0.211, +0.291]	+0.251	0	0	0.293	✓	-	2.11×10^{-9}	✓
CausalRCA vs max- Z	3	-0.248 [-0.304, -0.192]	-0.248	1	1	0.342	✓	-	1.32×10^{-13}	
CausalRCA vs RCD	3	-0.030 [-0.077, +0.017]	-0.030	2	0	0.159	✓	-	0.004	
CD-1min vs CIRCA	3	+0.077 [+0.021, +0.133]	+0.077	1	0	0.137	✓	-	0.002	
CD-1min vs ϵ -Diagnosis	3	+0.587 [+0.544, +0.632]	+0.587	0	0	0.269		-	0.880	✓
CD-1min vs max- Z	3	+0.088 [+0.032, +0.147]	+0.088	1	1	0.202	✓	-	1.68×10^{-5}	
CD-1min vs RCD	3	+0.306 [+0.250, +0.362]	+0.306	0	0	0.202		-	9.51×10^{-5}	
CIRCA vs ϵ -Diagnosis	3	+0.509 [+0.461, +0.560]	+0.509	0	0	0.162		-	0.047	✓
CIRCA vs max- Z	3	+0.011 [-0.051, +0.072]	+0.011	2	0	0.139	✓	-	0.055	
CIRCA vs RCD	3	+0.228 [+0.167, +0.286]	+0.228	0	0	0.093		-	0.172	
ϵ -Diagnosis vs max- Z	3	-0.499 [-0.552, -0.443]	-0.499	0	0	0.091		-	0.131	✓
ϵ -Diagnosis vs RCD	3	-0.281 [-0.329, -0.234]	-0.281	0	0	0.213		-	0.002	✓
max- Z vs RCD	3	+0.218 [+0.158, +0.275]	+0.218	1	0	0.221	✓	-	1.40×10^{-6}	
<i>RCAEval RE2, Published</i>										
BARO vs CausalRCA	3	+0.105 [+0.047, +0.163]	+0.105	2	1	0.307	✓	-	4.11×10^{-7}	
BARO vs CIRCA	3	-0.370 [-0.433, -0.307]	-0.370	1	1	0.437	✓	-	0	
BARO vs ϵ -Diagnosis	3	+0.230 [+0.174, +0.285]	+0.230	1	0	0.391	✓	-	3.33×10^{-14}	✓
BARO vs MicroRank	2	+0.344 [+0.267, +0.422]	+0.344	0	0	0.283		-	1.000	✓
BARO vs BARO (multi-source)	3	-0.015 [-0.041, +0.007]	-0.015	0	0	0.013		-	1.000	
BARO vs SimpleRCA	3	-0.211 [-0.270, -0.156]	-0.211	0	0	0.099		-	1.000	
BARO vs TraceRCA	2	+0.089 [+0.000, +0.172]	+0.089	0	0	0.063		-	1.000	
CausalRCA vs CIRCA	3	-0.475 [-0.538, -0.412]	-0.475	0	0	0.147		-	0.152	
CausalRCA vs ϵ -Diagnosis	3	+0.125 [+0.078, +0.172]	+0.125	1	0	0.116		-	2.63×10^{-4}	✓
CausalRCA vs MicroRank	2	+0.146 [+0.089, +0.204]	+0.146	0	0	0.086		-	0.004	✓
CausalRCA vs BARO (multi-source)	3	-0.120 [-0.179, -0.062]	-0.120	2	1	0.314	✓	-	3.29×10^{-7}	
CausalRCA vs SimpleRCA	3	-0.316 [-0.375, -0.256]	-0.316	0	0	0.253		-	0.002	
CausalRCA vs TraceRCA	2	-0.109 [-0.183, -0.035]	-0.109	1	1	0.306	✓	-	2.30×10^{-4}	
CIRCA vs ϵ -Diagnosis	3	+0.600 [+0.537, +0.663]	+0.600	0	0	0.048		-	0.152	✓
CIRCA vs MicroRank	2	+0.544 [+0.467, +0.622]	+0.544	0	0	0.173		-	2.63×10^{-4}	✓
CIRCA vs BARO (multi-source)	3	+0.356 [+0.289, +0.422]	+0.356	1	1	0.446	✓	-	0	
CIRCA vs SimpleRCA	3	+0.159 [+0.089, +0.230]	+0.159	1	1	0.396	✓	-	1.70×10^{-12}	
CIRCA vs TraceRCA	2	+0.289 [+0.189, +0.389]	+0.289	0	0	0.393	✓	-	6.98×10^{-8}	
ϵ -Diagnosis vs MicroRank	2	-0.044 [-0.083, -0.006]	-0.044	1	0	0.110	✓	-	0.002	✓
ϵ -Diagnosis vs BARO (multi-source)	3	-0.244 [-0.300, -0.189]	-0.244	1	0	0.401	✓	-	1.15×10^{-14}	✓
ϵ -Diagnosis vs SimpleRCA	3	-0.441 [-0.500, -0.381]	-0.441	0	0	0.356		-	9.09×10^{-12}	✓
ϵ -Diagnosis vs TraceRCA	2	-0.300 [-0.361, -0.239]	-0.300	0	0	0.330	✓	-	0.003	✓
MicroRank vs BARO (multi-source)	2	-0.367 [-0.444, -0.289]	-0.367	0	0	0.283		-	1.000	✓
MicroRank vs SimpleRCA	2	-0.572 [-0.650, -0.494]	-0.572	0	0	0.149		-	0.370	✓
MicroRank vs TraceRCA	2	-0.256 [-0.322, -0.189]	-0.256	0	0	0.220		-	1.000	✓
BARO (multi-source) vs SimpleRCA	3	-0.196 [-0.256, -0.137]	-0.196	0	0	0.096		-	1.000	
BARO (multi-source) vs TraceRCA	2	+0.111 [+0.022, +0.200]	+0.111	0	0	0.063		-	1.000	
SimpleRCA vs TraceRCA	2	+0.317 [+0.228, +0.411]	+0.317	0	0	0.071		-	1.000	
<i>RCAEval RE2, All</i>										
Alert-count vs BARO	3	-0.022 [-0.089, +0.044]	-0.022	2	1	0.396	✓	-	4.17×10^{-14}	
Alert-count vs CausalRCA	3	+0.083 [+0.022, +0.147]	+0.083	0	0	0.137	✓	-	0.258	
Alert-count vs CD-1min	3	-0.552 [-0.615, -0.485]	-0.552	0	0	0.028		-	1.000	
Alert-count vs CIRCA	3	-0.393 [-0.467, -0.319]	-0.393	0	0	0.061		-	1.000	
Alert-count vs ϵ -Diagnosis	3	+0.207 [+0.152, +0.267]	+0.207	0	0	0.028		-	0.182	✓
Alert-count vs max- Z	3	-0.407 [-0.481, -0.330]	-0.407	0	0	0.188		-	0.068	
Alert-count vs MicroRank	2	+0.150 [+0.089, +0.217]	+0.150	0	0	0.086		-	0.008	✓
Alert-count vs BARO (multi-source)	3	-0.037 [-0.104, +0.033]	-0.037	2	1	0.406	✓	-	1.11×10^{-14}	
Alert-count vs SimpleRCA	3	-0.233 [-0.304, -0.159]	-0.233	1	1	0.367	✓	-	2.07×10^{-10}	
Alert-count vs TraceRCA	2	-0.106 [-0.183, -0.028]	-0.106	1	1	0.306	✓	-	3.82×10^{-4}	
BARO vs CausalRCA	3	+0.105 [+0.047, +0.163]	+0.105	2	1	0.307	✓	-	7.40×10^{-7}	
BARO vs CD-1min	3	-0.530 [-0.585, -0.474]	-0.530	0	0	0.381		-	1.83×10^{-12}	
BARO vs CIRCA	3	-0.370 [-0.433, -0.307]	-0.370	1	1	0.437	✓	-	0	
BARO vs ϵ -Diagnosis	3	+0.230 [+0.174, +0.285]	+0.230	1	0	0.391	✓	-	6.13×10^{-14}	✓
BARO vs max- Z	3	-0.385 [-0.444, -0.322]	-0.385	1	1	0.560	✓	-	0	
BARO vs MicroRank	2	+0.344 [+0.267, +0.422]	+0.344	0	0	0.283		-	1.000	✓
BARO vs BARO (multi-source)	3	-0.015 [-0.041, +0.007]	-0.015	0	0	0.013		-	1.000	
BARO vs SimpleRCA	3	-0.211 [-0.270, -0.156]	-0.211	0	0	0.099		-	1.000	
BARO vs TraceRCA	2	+0.089 [+0.000, +0.172]	+0.089	0	0	0.063		-	1.000	

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Table 8: All method pairs in the main analysis (seed-averaged). Pooled Δ (equal) weights subsystems equally and is shown with its 95% CI; the case-weighted pooled Δ weights subsystems by case count. #rev is the number of subsystems whose Δ has the opposite sign of the equal-weight pooled Δ ; #rev CI counts those whose CI excludes 0. Below line marks $|\text{pooled } \Delta| < \text{SD of subsystem } \Delta$. The 95% prediction interval is reported only where the random-effects fit supplies one. LRT p is Holm-adjusted. Sep. marks complete separation.

Pair	k	Pooled Δ (equal) [CI]	Pooled Δ (case-wtd.)	#rev	#rev CI	SD of Δ_s	Below line	95% PI	LRT p (Holm)	Sep.
CausalRCA vs CD-1min	3	-0.635 [-0.686, -0.581]	-0.635	0	0	0.143		- 0.014		
CausalRCA vs CIRCA	3	-0.475 [-0.538, -0.412]	-0.475	0	0	0.147		- 0.240		
CausalRCA vs ϵ -Diagnosis	3	+0.125 [+0.078, +0.172]	+0.125	1	0	0.116		- 4.67 $\times 10^{-4}$		✓
CausalRCA vs max- Z	3	-0.490 [-0.549, -0.431]	-0.490	0	0	0.256		- 3.43 $\times 10^{-4}$		
CausalRCA vs MicroRank	2	+0.146 [+0.089, +0.204]	+0.146	0	0	0.086		- 0.008		✓
CausalRCA vs BARO (multi-source)	3	-0.120 [-0.179, -0.062]	-0.120	2	1	0.314	✓	- 5.80 $\times 10^{-7}$		
CausalRCA vs SimpleRCA	3	-0.316 [-0.375, -0.256]	-0.316	0	0	0.253		- 0.004		
CausalRCA vs TraceRCA	2	-0.109 [-0.183, -0.035]	-0.109	1	1	0.306	✓	- 3.99 $\times 10^{-4}$		
CD-1min vs CIRCA	3	+0.159 [+0.100, +0.219]	+0.159	0	0	0.089		- 0.833		
CD-1min vs ϵ -Diagnosis	3	+0.759 [+0.704, +0.811]	+0.759	0	0	0.051		- 0.832		✓
CD-1min vs max- Z	3	+0.144 [+0.089, +0.200]	+0.144	1	0	0.215	✓	- 0.007		
CD-1min vs MicroRank	2	+0.700 [+0.622, +0.778]	+0.700	0	0	0.047		- 0.023		✓
CD-1min vs BARO (multi-source)	3	+0.515 [+0.456, +0.570]	+0.515	0	0	0.391		- 5.50 $\times 10^{-13}$		
CD-1min vs SimpleRCA	3	+0.319 [+0.259, +0.378]	+0.319	1	0	0.358	✓	- 2.07 $\times 10^{-10}$		
CD-1min vs TraceRCA	2	+0.444 [+0.356, +0.533]	+0.444	0	0	0.267		- 0.003		
CIRCA vs ϵ -Diagnosis	3	+0.600 [+0.537, +0.663]	+0.600	0	0	0.048		- 0.240		✓
CIRCA vs max- Z	3	-0.015 [-0.078, +0.048]	-0.015	1	0	0.130	✓	- 0.445		
CIRCA vs MicroRank	2	+0.544 [+0.467, +0.622]	+0.544	0	0	0.173		- 4.67 $\times 10^{-4}$		✓
CIRCA vs BARO (multi-source)	3	+0.356 [+0.289, +0.422]	+0.356	1	1	0.446	✓	- 0		
CIRCA vs SimpleRCA	3	+0.159 [+0.089, +0.230]	+0.159	1	1	0.396	✓	- 3.05 $\times 10^{-12}$		
CIRCA vs TraceRCA	2	+0.289 [+0.189, +0.389]	+0.289	0	0	0.393	✓	- 1.24 $\times 10^{-7}$		
ϵ -Diagnosis vs max- Z	3	-0.615 [-0.674, -0.552]	-0.615	0	0	0.179		- 0.833		✓
ϵ -Diagnosis vs MicroRank	2	-0.044 [-0.083, -0.006]	-0.044	1	0	0.110	✓	- 0.003		✓
ϵ -Diagnosis vs BARO (multi-source)	3	-0.244 [-0.300, -0.189]	-0.244	1	0	0.401	✓	- 2.13 $\times 10^{-14}$		✓
ϵ -Diagnosis vs SimpleRCA	3	-0.441 [-0.500, -0.381]	-0.441	0	0	0.356		- 1.66 $\times 10^{-11}$		✓
ϵ -Diagnosis vs TraceRCA	2	-0.300 [-0.361, -0.239]	-0.300	0	0	0.330	✓	- 0.007		✓
max- Z vs MicroRank	2	+0.533 [+0.456, +0.611]	+0.533	0	0	0.346		- 4.14 $\times 10^{-7}$		✓
max- Z vs BARO (multi-source)	3	+0.370 [+0.307, +0.433]	+0.370	1	1	0.567	✓	- 0		
max- Z vs SimpleRCA	3	+0.174 [+0.111, +0.237]	+0.174	1	1	0.508	✓	- 0		
max- Z vs TraceRCA	2	+0.278 [+0.183, +0.372]	+0.278	1	0	0.566	✓	- 1.11 $\times 10^{-14}$		
MicroRank vs BARO (multi-source)	2	-0.367 [-0.444, -0.289]	-0.367	0	0	0.283		- 1.000		✓
MicroRank vs SimpleRCA	2	-0.572 [-0.650, -0.494]	-0.572	0	0	0.149		- 0.555		✓
MicroRank vs TraceRCA	2	-0.256 [-0.322, -0.189]	-0.256	0	0	0.220		- 1.000		✓
BARO (multi-source) vs SimpleRCA	3	-0.196 [-0.256, -0.137]	-0.196	0	0	0.096		- 1.000		
BARO (multi-source) vs TraceRCA	2	+0.111 [+0.022, +0.200]	+0.111	0	0	0.063		- 1.000		
SimpleRCA vs TraceRCA	2	+0.317 [+0.228, +0.411]	+0.317	0	0	0.071		- 1.000		
<i>OpenRCA, Published</i>										
BARO vs ϵ -Diagnosis	4	+0.086 [+0.054, +0.118]	+0.090	0	0	0.055		[-0.153, +0.311]	1.000	
BARO vs RCD	4	+0.027 [+0.002, +0.051]	+0.038	1	0	0.036	✓	[-0.109, +0.157]	1.000	
ϵ -Diagnosis vs RCD	4	-0.059 [-0.089, -0.031]	-0.052	0	0	0.061	✓	[-0.281, +0.181]	1.000	
<i>OpenRCA, All</i>										
Alert-count vs BARO	4	+0.062 [+0.030, +0.094]	+0.053	0	0	0.036		[-0.052, +0.171]	1.000	
Alert-count vs CD-1min	4	+0.018 [-0.029, +0.064]	+0.012	1	0	0.042	✓	[-0.087, +0.097]	1.000	
Alert-count vs ϵ -Diagnosis	4	+0.148 [+0.113, +0.185]	+0.143	0	0	0.038		[+0.064, +0.205]	1.000	
Alert-count vs max- Z	4	+0.009 [-0.028, +0.046]	+0.016	2	0	0.027	✓	[-0.060, +0.082]	1.000	
Alert-count vs RCD	4	+0.089 [+0.055, +0.122]	+0.091	0	0	0.033		[+0.017, +0.158]	1.000	
BARO vs CD-1min	4	-0.044 [-0.087, -0.003]	-0.041	1	0	0.061	✓	[-0.270, +0.169]	1.000	
BARO vs ϵ -Diagnosis	4	+0.086 [+0.054, +0.118]	+0.090	0	0	0.055		[-0.153, +0.311]	1.000	
BARO vs max- Z	4	-0.053 [-0.090, -0.018]	-0.037	1	0	0.063	✓	[-0.324, +0.221]	1.000	
BARO vs RCD	4	+0.027 [+0.002, +0.051]	+0.038	1	0	0.036	✓	[-0.109, +0.157]	1.000	
CD-1min vs ϵ -Diagnosis	4	+0.130 [+0.090, +0.171]	+0.131	0	0	0.014		[+0.050, +0.210]	1.000	
CD-1min vs max- Z	4	-0.009 [-0.055, +0.037]	+0.003	2	0	0.044	✓	[-0.088, +0.096]	1.000	
CD-1min vs RCD	4	+0.071 [+0.033, +0.110]	+0.079	1	0	0.070		[-0.153, +0.319]	1.000	
ϵ -Diagnosis vs max- Z	4	-0.139 [-0.177, -0.103]	-0.127	0	0	0.041		[-0.230, -0.020]	1.000	
ϵ -Diagnosis vs RCD	4	-0.059 [-0.089, -0.031]	-0.052	0	0	0.061	✓	[-0.281, +0.181]	1.000	
max- Z vs RCD	4	+0.080 [+0.043, +0.116]	+0.075	0	0	0.049		[-0.096, +0.251]	1.000	
<i>PetShop, Published</i>										
BARO vs CIRCA	4	-0.243 [-0.418, -0.058]	-0.206	0	0	0.106		[-0.442, +0.083]	1.000	
BARO vs Counterfactual	4	-0.007 [-0.163, +0.147]	-0.044	1	0	0.114	✓	[-0.343, +0.241]	1.000	
BARO vs ϵ -Diagnosis	4	+0.171 [+0.036, +0.312]	+0.162	0	0	0.055		[-0.044, +0.366]	1.000	✓
BARO vs RCD	4	-0.135 [-0.300, +0.041]	-0.074	1	1	0.219	✓	[-0.913, +0.775]	0.173	✓
CIRCA vs Counterfactual	4	+0.236 [+0.055, +0.404]	+0.162	1	0	0.195		[-0.497, +0.861]	1.000	
CIRCA vs ϵ -Diagnosis	4	+0.413 [+0.274, +0.550]	+0.368	0	0	0.110		[+0.103, +0.609]	1.000	✓
CIRCA vs RCD	4	+0.108 [-0.089, +0.298]	+0.132	1	0	0.187	✓	[-0.153, +0.523]	0.233	✓
Counterfactual vs ϵ -Diagnosis	4	+0.178 [+0.087, +0.288]	+0.206	0	0	0.088		[-0.019, +0.392]	1.000	✓
Counterfactual vs RCD	4	-0.127 [-0.300, +0.058]	-0.029	1	1	0.333	✓	[-1.493, +1.338]	0.002	✓
ϵ -Diagnosis vs RCD	4	-0.305 [-0.452, -0.151]	-0.235	0	0	0.258		- 1.000		✓
<i>PetShop, All</i>										
Alert-count vs BARO	4	-0.130 [-0.262, -0.005]	-0.132	0	0	0.103		[-0.354, +0.077]	1.000	✓
Alert-count vs Traversal	4	-0.543 [-0.675, -0.401]	-0.500	0	0	0.133		[-0.870, -0.156]	1.000	✓
Alert-count vs CD-1min	4	+0.000 [-0.125, +0.115]	+0.000	0	0	0.107	✓	[-0.142, +0.133]	1.000	✓

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Table 8: All method pairs in the main analysis (seed-averaged). Pooled Δ (equal) weights subsystems equally and is shown with its 95% CI; the case-weighted pooled Δ weights subsystems by case count. #rev is the number of subsystems whose Δ has the opposite sign of the equal-weight pooled Δ ; #rev CI counts those whose CI excludes 0. Below line marks $|\text{pooled } \Delta| < \text{SD of subsystem } \Delta$. The 95% prediction interval is reported only where the random-effects fit supplies one. LRT p is Holm-adjusted. Sep. marks complete separation.

Pair	k	Pooled Δ (equal) [CI]	Pooled Δ (case-wtd.)	#rev	#rev CI	SD of Δ_s	Below line	95% PI	LRT p (Holm)	Sep.
Alert-count vs CIRCA	4	-0.373 [-0.529, -0.202]	-0.338	0	0	0.110		[-0.610, -0.062]	1.000	✓
Alert-count vs Counterfactual	4	-0.137 [-0.291, +0.010]	-0.176	0	0	0.111		[-0.398, +0.054]	1.000	✓
Alert-count vs ϵ -Diagnosis	4	+0.041 [-0.062, +0.144]	+0.029	0	0	0.059	✓	-	1.000	✓
Alert-count vs max- Z	4	-0.183 [-0.327, -0.038]	-0.147	0	0	0.143		[-0.509, +0.236]	1.000	✓
Alert-count vs Random walk	4	+0.041 [-0.062, +0.135]	+0.029	1	0	0.156	✓	-	1.000	✓
Alert-count vs Ranked correlation	4	-0.567 [-0.692, -0.440]	-0.603	0	0	0.223		[-1.392, +0.234]	1.000	✓
Alert-count vs RCD	4	-0.264 [-0.428, -0.096]	-0.206	1	0	0.287	✓	[-1.494, +0.998]	0.509	✓
BARO vs Traversal	4	-0.413 [-0.555, -0.269]	-0.368	0	0	0.163		[-0.769, -0.008]	1.000	
BARO vs CD-1min	4	+0.130 [+0.048, +0.231]	+0.132	0	0	0.017		[-0.050, +0.312]	1.000	✓
BARO vs CIRCA	4	-0.243 [-0.418, -0.058]	-0.206	0	0	0.106		[-0.442, +0.083]	1.000	
BARO vs Counterfactual	4	-0.007 [-0.163, +0.147]	-0.044	1	0	0.114	✓	[-0.343, +0.241]	1.000	
BARO vs ϵ -Diagnosis	4	+0.171 [+0.036, +0.312]	+0.162	0	0	0.055		[-0.044, +0.366]	1.000	✓
BARO vs max- Z	4	-0.053 [-0.147, +0.019]	-0.015	1	0	0.096	✓	[-0.306, +0.272]	1.000	
BARO vs Random walk	4	+0.171 [+0.036, +0.312]	+0.162	0	0	0.055		[-0.044, +0.366]	1.000	✓
BARO vs Ranked correlation	4	-0.438 [-0.579, -0.298]	-0.471	0	0	0.140		[-0.749, -0.149]	1.000	
BARO vs RCD	4	-0.135 [-0.300, +0.041]	-0.074	1	1	0.219	✓	[-0.913, +0.775]	0.997	✓
Traversal vs CD-1min	4	+0.543 [+0.406, +0.673]	+0.500	0	0	0.153		[+0.087, +0.938]	1.000	✓
Traversal vs CIRCA	4	+0.171 [+0.017, +0.322]	+0.162	0	0	0.063		[-0.137, +0.428]	1.000	
Traversal vs Counterfactual	4	+0.406 [+0.255, +0.548]	+0.324	0	0	0.235		[-0.538, +1.308]	1.000	
Traversal vs ϵ -Diagnosis	4	+0.584 [+0.450, +0.709]	+0.529	0	0	0.152		[+0.057, +1.036]	1.000	✓
Traversal vs max- Z	4	+0.361 [+0.221, +0.500]	+0.353	0	0	0.108		[+0.057, +0.646]	1.000	
Traversal vs Random walk	4	+0.584 [+0.459, +0.704]	+0.529	0	0	0.209		[-0.288, +1.453]	1.000	✓
Traversal vs Ranked correlation	4	-0.024 [-0.185, +0.137]	-0.103	1	1	0.302	✓	[-1.298, +1.231]	1.000	
Traversal vs RCD	4	+0.279 [+0.079, +0.464]	+0.294	0	0	0.216		[+0.005, +0.638]	1.000	✓
CD-1min vs CIRCA	4	-0.373 [-0.548, -0.178]	-0.338	0	0	0.096		[-0.595, -0.037]	1.000	✓
CD-1min vs Counterfactual	4	-0.137 [-0.264, -0.024]	-0.176	0	0	0.127		[-0.490, +0.150]	1.000	✓
CD-1min vs ϵ -Diagnosis	4	+0.041 [-0.062, +0.154]	+0.029	0	0	0.059	✓	-	1.000	✓
CD-1min vs max- Z	4	-0.183 [-0.298, -0.079]	-0.147	0	0	0.084		[-0.295, +0.051]	1.000	✓
CD-1min vs Random walk	4	+0.041 [-0.062, +0.154]	+0.029	0	0	0.059	✓	-	1.000	✓
CD-1min vs Ranked correlation	4	-0.567 [-0.702, -0.433]	-0.603	0	0	0.150		[-1.021, -0.189]	1.000	✓
CD-1min vs RCD	4	-0.264 [-0.397, -0.139]	-0.206	0	0	0.206		-	1.000	✓
CIRCA vs Counterfactual	4	+0.236 [+0.055, +0.404]	+0.162	1	0	0.195		[-0.497, +0.861]	1.000	
CIRCA vs ϵ -Diagnosis	4	+0.413 [+0.274, +0.550]	+0.368	0	0	0.110		[+0.103, +0.609]	1.000	✓
CIRCA vs max- Z	4	+0.190 [+0.019, +0.363]	+0.191	0	0	0.051		[-0.097, +0.478]	1.000	
CIRCA vs Random walk	4	+0.413 [+0.281, +0.546]	+0.368	0	0	0.150		[+0.057, +0.665]	1.000	✓
CIRCA vs Ranked correlation	4	-0.195 [-0.399, +0.007]	-0.265	1	0	0.243	✓	[-0.948, +0.400]	1.000	
CIRCA vs RCD	4	+0.108 [-0.089, +0.298]	+0.132	1	0	0.187	✓	[-0.153, +0.523]	1.000	✓
Counterfactual vs ϵ -Diagnosis	4	+0.178 [+0.087, +0.288]	+0.206	0	0	0.088		[-0.019, +0.392]	1.000	✓
Counterfactual vs max- Z	4	-0.046 [-0.212, +0.132]	+0.029	1	1	0.204	✓	[-0.681, +0.729]	1.000	
Counterfactual vs Random walk	4	+0.178 [+0.053, +0.312]	+0.206	0	0	0.135		[-0.022, +0.422]	1.000	✓
Counterfactual vs Ranked correlation	4	-0.430 [-0.606, -0.236]	-0.426	0	0	0.153		[-0.731, -0.116]	1.000	
Counterfactual vs RCD	4	-0.127 [-0.300, +0.058]	-0.029	1	1	0.333	✓	[-1.493, +1.338]	0.010	✓
ϵ -Diagnosis vs max- Z	4	-0.224 [-0.370, -0.070]	-0.176	0	0	0.124		[-0.420, +0.135]	1.000	✓
ϵ -Diagnosis vs Random walk	4	+0.000 [-0.094, +0.094]	+0.000	0	0	0.102	✓	-	1.000	✓
ϵ -Diagnosis vs Ranked correlation	4	-0.608 [-0.757, -0.445]	-0.632	0	0	0.174		[-0.916, -0.402]	1.000	✓
ϵ -Diagnosis vs RCD	4	-0.305 [-0.452, -0.151]	-0.235	0	0	0.258		-	1.000	✓
max- Z vs Random walk	4	+0.224 [+0.077, +0.368]	+0.176	0	0	0.124		[-0.135, +0.420]	1.000	✓
max- Z vs Ranked correlation	4	-0.385 [-0.514, -0.264]	-0.456	0	0	0.216		[-1.307, +0.526]	1.000	
max- Z vs RCD	4	-0.082 [-0.236, +0.072]	-0.059	1	0	0.148	✓	[-0.444, +0.412]	1.000	✓
Random walk vs Ranked correlation	4	-0.608 [-0.743, -0.466]	-0.632	0	0	0.097		[-0.896, -0.389]	1.000	✓
Random walk vs RCD	4	-0.305 [-0.438, -0.180]	-0.235	0	0	0.214		-	1.000	✓
Ranked correlation vs RCD	4	+0.303 [+0.139, +0.466]	+0.397	0	0	0.290		[-0.940, +1.631]	0.024	✓

Per-seed pair results

The same pair contrasts computed separately on each of the three seeds. Source: [paper/data/rq1/pairs.csv](#).

Table 9: Per-seed pair results (seeds 0, 1, and 2). Pooled Δ is the equal-subsystem-weight difference with its 95% CI. #rev is the number of subsystems reversing the pooled sign; the parenthetical count is the number of those reversals whose CI excludes 0.

Family	Layer	Pair	Seed	Pooled Δ [CI]	#rev (#rev CI)
RCAEval RE1	Published	BARO vs CausalRCA	0	+0.363 [+0.307, +0.419]	0 (0)
RCAEval RE1	Published	BARO vs CausalRCA	1	+0.405 [+0.349, +0.459]	0 (0)
RCAEval RE1	Published	BARO vs CausalRCA	2	+0.448 [+0.397, +0.499]	0 (0)
RCAEval RE1	Published	BARO vs RCD	0	+0.389 [+0.333, +0.445]	0 (0)
RCAEval RE1	Published	BARO vs RCD	1	+0.371 [+0.312, +0.429]	0 (0)
RCAEval RE1	Published	BARO vs RCD	2	+0.365 [+0.307, +0.424]	0 (0)
RCAEval RE1	Published	CausalRCA vs CIRCA	0	-0.216 [-0.275, -0.157]	0 (0)
RCAEval RE1	Published	CausalRCA vs CIRCA	1	-0.259 [-0.315, -0.203]	0 (0)

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Table 9: Per-seed pair results (seeds 0, 1, and 2). Pooled Δ is the equal-subsystem-weight difference with its 95% CI. #rev is the number of subsystems reversing the pooled sign; the parenthetical count is the number of those reversals whose CI excludes 0.

Family	Layer	Pair	Seed	Pooled Δ [CI]	#rev (#rev CI)
RCAEval RE1	Published	CausalRCA vs CIRCA	2	-0.301 [-0.352, -0.248]	0 (0)
RCAEval RE1	Published	CausalRCA vs ϵ -Diagnosis	0	+0.293 [+0.243, +0.344]	0 (0)
RCAEval RE1	Published	CausalRCA vs ϵ -Diagnosis	1	+0.251 [+0.205, +0.299]	0 (0)
RCAEval RE1	Published	CausalRCA vs ϵ -Diagnosis	2	+0.208 [+0.165, +0.251]	1 (0)
RCAEval RE1	Published	CausalRCA vs RCD	0	+0.027 [-0.029, +0.083]	1 (1)
RCAEval RE1	Published	CausalRCA vs RCD	1	-0.035 [-0.091, +0.021]	1 (1)
RCAEval RE1	Published	CausalRCA vs RCD	2	-0.083 [-0.136, -0.029]	1 (0)
RCAEval RE1	Published	CIRCA vs RCD	0	+0.243 [+0.181, +0.304]	0 (0)
RCAEval RE1	Published	CIRCA vs RCD	1	+0.224 [+0.160, +0.285]	0 (0)
RCAEval RE1	Published	CIRCA vs RCD	2	+0.219 [+0.155, +0.283]	0 (0)
RCAEval RE1	Published	ϵ -Diagnosis vs RCD	0	-0.267 [-0.317, -0.219]	0 (0)
RCAEval RE1	Published	ϵ -Diagnosis vs RCD	1	-0.285 [-0.336, -0.237]	0 (0)
RCAEval RE1	Published	ϵ -Diagnosis vs RCD	2	-0.291 [-0.341, -0.243]	0 (0)
RCAEval RE1	All	Alert-count vs CausalRCA	0	-0.080 [-0.139, -0.021]	1 (1)
RCAEval RE1	All	Alert-count vs CausalRCA	1	-0.037 [-0.093, +0.019]	1 (1)
RCAEval RE1	All	Alert-count vs CausalRCA	2	+0.005 [-0.045, +0.056]	1 (1)
RCAEval RE1	All	Alert-count vs RCD	0	-0.053 [-0.112, +0.005]	0 (0)
RCAEval RE1	All	Alert-count vs RCD	1	-0.072 [-0.128, -0.016]	0 (0)
RCAEval RE1	All	Alert-count vs RCD	2	-0.077 [-0.136, -0.019]	0 (0)
RCAEval RE1	All	BARO vs CausalRCA	0	+0.363 [+0.307, +0.419]	0 (0)
RCAEval RE1	All	BARO vs CausalRCA	1	+0.405 [+0.349, +0.459]	0 (0)
RCAEval RE1	All	BARO vs CausalRCA	2	+0.448 [+0.397, +0.499]	0 (0)
RCAEval RE1	All	BARO vs RCD	0	+0.389 [+0.333, +0.445]	0 (0)
RCAEval RE1	All	BARO vs RCD	1	+0.371 [+0.312, +0.429]	0 (0)
RCAEval RE1	All	BARO vs RCD	2	+0.365 [+0.307, +0.424]	0 (0)
RCAEval RE1	All	CausalRCA vs CD-1min	0	-0.293 [-0.349, -0.240]	0 (0)
RCAEval RE1	All	CausalRCA vs CD-1min	1	-0.336 [-0.392, -0.283]	0 (0)
RCAEval RE1	All	CausalRCA vs CD-1min	2	-0.379 [-0.427, -0.331]	0 (0)
RCAEval RE1	All	CausalRCA vs CIRCA	0	-0.216 [-0.275, -0.157]	0 (0)
RCAEval RE1	All	CausalRCA vs CIRCA	1	-0.259 [-0.315, -0.203]	0 (0)
RCAEval RE1	All	CausalRCA vs CIRCA	2	-0.301 [-0.352, -0.248]	0 (0)
RCAEval RE1	All	CausalRCA vs ϵ -Diagnosis	0	+0.293 [+0.243, +0.344]	0 (0)
RCAEval RE1	All	CausalRCA vs ϵ -Diagnosis	1	+0.251 [+0.205, +0.299]	0 (0)
RCAEval RE1	All	CausalRCA vs ϵ -Diagnosis	2	+0.208 [+0.165, +0.251]	1 (0)
RCAEval RE1	All	CausalRCA vs max- Z	0	-0.205 [-0.267, -0.144]	1 (1)
RCAEval RE1	All	CausalRCA vs max- Z	1	-0.248 [-0.309, -0.184]	1 (0)
RCAEval RE1	All	CausalRCA vs max- Z	2	-0.291 [-0.347, -0.235]	1 (1)
RCAEval RE1	All	CausalRCA vs RCD	0	+0.027 [-0.029, +0.083]	1 (1)
RCAEval RE1	All	CausalRCA vs RCD	1	-0.035 [-0.091, +0.021]	1 (1)
RCAEval RE1	All	CausalRCA vs RCD	2	-0.083 [-0.136, -0.029]	1 (0)
RCAEval RE1	All	CD-1min vs RCD	0	+0.320 [+0.261, +0.379]	0 (0)
RCAEval RE1	All	CD-1min vs RCD	1	+0.301 [+0.243, +0.360]	0 (0)
RCAEval RE1	All	CD-1min vs RCD	2	+0.296 [+0.235, +0.357]	0 (0)
RCAEval RE1	All	CIRCA vs RCD	0	+0.243 [+0.181, +0.304]	0 (0)
RCAEval RE1	All	CIRCA vs RCD	1	+0.224 [+0.160, +0.285]	0 (0)
RCAEval RE1	All	CIRCA vs RCD	2	+0.219 [+0.155, +0.283]	0 (0)
RCAEval RE1	All	ϵ -Diagnosis vs RCD	0	-0.267 [-0.317, -0.219]	0 (0)
RCAEval RE1	All	ϵ -Diagnosis vs RCD	1	-0.285 [-0.336, -0.237]	0 (0)
RCAEval RE1	All	ϵ -Diagnosis vs RCD	2	-0.291 [-0.341, -0.243]	0 (0)
RCAEval RE1	All	max- Z vs RCD	0	+0.232 [+0.171, +0.291]	1 (0)
RCAEval RE1	All	max- Z vs RCD	1	+0.213 [+0.149, +0.275]	1 (0)
RCAEval RE1	All	max- Z vs RCD	2	+0.208 [+0.144, +0.269]	1 (0)
RCAEval RE2	Published	BARO vs CausalRCA	0	+0.063 [-0.004, +0.130]	2 (1)
RCAEval RE2	Published	BARO vs CausalRCA	1	+0.137 [+0.074, +0.200]	2 (1)
RCAEval RE2	Published	BARO vs CausalRCA	2	+0.115 [+0.052, +0.178]	2 (0)
RCAEval RE2	Published	CausalRCA vs CIRCA	0	-0.433 [-0.504, -0.363]	0 (0)
RCAEval RE2	Published	CausalRCA vs CIRCA	1	-0.507 [-0.574, -0.441]	0 (0)
RCAEval RE2	Published	CausalRCA vs CIRCA	2	-0.485 [-0.556, -0.415]	0 (0)
RCAEval RE2	Published	CausalRCA vs ϵ -Diagnosis	0	+0.167 [+0.111, +0.222]	1 (0)
RCAEval RE2	Published	CausalRCA vs ϵ -Diagnosis	1	+0.093 [+0.041, +0.144]	0 (0)
RCAEval RE2	Published	CausalRCA vs ϵ -Diagnosis	2	+0.115 [+0.059, +0.170]	1 (0)
RCAEval RE2	Published	CausalRCA vs MicroRank	0	+0.211 [+0.139, +0.283]	0 (0)
RCAEval RE2	Published	CausalRCA vs MicroRank	1	+0.083 [+0.022, +0.150]	0 (0)
RCAEval RE2	Published	CausalRCA vs MicroRank	2	+0.144 [+0.078, +0.211]	0 (0)
RCAEval RE2	Published	CausalRCA vs BARO (multi-source)	0	-0.078 [-0.144, -0.011]	2 (0)
RCAEval RE2	Published	CausalRCA vs BARO (multi-source)	1	-0.152 [-0.215, -0.089]	1 (1)
RCAEval RE2	Published	CausalRCA vs BARO (multi-source)	2	-0.130 [-0.193, -0.067]	2 (0)
RCAEval RE2	Published	CausalRCA vs SimpleRCA	0	-0.274 [-0.344, -0.204]	0 (0)
RCAEval RE2	Published	CausalRCA vs SimpleRCA	1	-0.348 [-0.411, -0.281]	0 (0)
RCAEval RE2	Published	CausalRCA vs SimpleRCA	2	-0.326 [-0.393, -0.263]	0 (0)
RCAEval RE2	Published	CausalRCA vs TraceRCA	0	-0.044 [-0.128, +0.039]	1 (1)
RCAEval RE2	Published	CausalRCA vs TraceRCA	1	-0.172 [-0.256, -0.089]	1 (0)
RCAEval RE2	Published	CausalRCA vs TraceRCA	2	-0.111 [-0.189, -0.033]	1 (0)
RCAEval RE2	All	Alert-count vs CausalRCA	0	+0.041 [-0.030, +0.115]	2 (0)
RCAEval RE2	All	Alert-count vs CausalRCA	1	+0.115 [+0.052, +0.181]	0 (0)
RCAEval RE2	All	Alert-count vs CausalRCA	2	+0.093 [+0.026, +0.163]	1 (0)
RCAEval RE2	All	BARO vs CausalRCA	0	+0.063 [-0.004, +0.130]	2 (1)
RCAEval RE2	All	BARO vs CausalRCA	1	+0.137 [+0.074, +0.200]	2 (1)
RCAEval RE2	All	BARO vs CausalRCA	2	+0.115 [+0.052, +0.178]	2 (0)

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Table 9: Per-seed pair results (seeds 0, 1, and 2). Pooled Δ is the equal-subsystem-weight difference with its 95% CI. #rev is the number of subsystems reversing the pooled sign; the parenthetical count is the number of those reversals whose CI excludes 0.

Family	Layer	Pair	Seed	Pooled Δ [CI]	#rev (#rev CI)
RCAEval RE2	All	CausalRCA vs CD-1min	0	-0.593 [-0.652, -0.533]	0 (0)
RCAEval RE2	All	CausalRCA vs CD-1min	1	-0.667 [-0.726, -0.607]	0 (0)
RCAEval RE2	All	CausalRCA vs CD-1min	2	-0.644 [-0.700, -0.585]	0 (0)
RCAEval RE2	All	CausalRCA vs CIRCA	0	-0.433 [-0.504, -0.363]	0 (0)
RCAEval RE2	All	CausalRCA vs CIRCA	1	-0.507 [-0.574, -0.441]	0 (0)
RCAEval RE2	All	CausalRCA vs CIRCA	2	-0.485 [-0.556, -0.415]	0 (0)
RCAEval RE2	All	CausalRCA vs ϵ -Diagnosis	0	+0.167 [+0.111, +0.222]	1 (0)
RCAEval RE2	All	CausalRCA vs ϵ -Diagnosis	1	+0.093 [+0.041, +0.144]	0 (0)
RCAEval RE2	All	CausalRCA vs ϵ -Diagnosis	2	+0.115 [+0.059, +0.170]	1 (0)
RCAEval RE2	All	CausalRCA vs max- Z	0	-0.448 [-0.519, -0.378]	0 (0)
RCAEval RE2	All	CausalRCA vs max- Z	1	-0.522 [-0.585, -0.459]	0 (0)
RCAEval RE2	All	CausalRCA vs max- Z	2	-0.500 [-0.563, -0.437]	0 (0)
RCAEval RE2	All	CausalRCA vs MicroRank	0	+0.211 [+0.139, +0.283]	0 (0)
RCAEval RE2	All	CausalRCA vs MicroRank	1	+0.083 [+0.022, +0.150]	0 (0)
RCAEval RE2	All	CausalRCA vs MicroRank	2	+0.144 [+0.078, +0.211]	0 (0)
RCAEval RE2	All	CausalRCA vs BARO (multi-source)	0	-0.078 [-0.144, -0.011]	2 (0)
RCAEval RE2	All	CausalRCA vs BARO (multi-source)	1	-0.152 [-0.215, -0.089]	1 (1)
RCAEval RE2	All	CausalRCA vs BARO (multi-source)	2	-0.130 [-0.193, -0.067]	2 (0)
RCAEval RE2	All	CausalRCA vs SimpleRCA	0	-0.274 [-0.344, -0.204]	0 (0)
RCAEval RE2	All	CausalRCA vs SimpleRCA	1	-0.348 [-0.411, -0.281]	0 (0)
RCAEval RE2	All	CausalRCA vs SimpleRCA	2	-0.326 [-0.393, -0.263]	0 (0)
RCAEval RE2	All	CausalRCA vs TraceRCA	0	-0.044 [-0.128, +0.039]	1 (1)
RCAEval RE2	All	CausalRCA vs TraceRCA	1	-0.172 [-0.256, -0.089]	1 (0)
RCAEval RE2	All	CausalRCA vs TraceRCA	2	-0.111 [-0.189, -0.033]	1 (0)
OpenRCA	Published	BARO vs RCD	0	+0.025 [+0.001, +0.049]	0 (0)
OpenRCA	Published	BARO vs RCD	1	+0.027 [-0.004, +0.057]	1 (0)
OpenRCA	Published	BARO vs RCD	2	+0.029 [+0.002, +0.057]	1 (0)
OpenRCA	Published	ϵ -Diagnosis vs RCD	0	-0.061 [-0.092, -0.032]	0 (0)
OpenRCA	Published	ϵ -Diagnosis vs RCD	1	-0.059 [-0.094, -0.026]	1 (0)
OpenRCA	Published	ϵ -Diagnosis vs RCD	2	-0.056 [-0.088, -0.027]	0 (0)
OpenRCA	All	Alert-count vs RCD	0	+0.087 [+0.054, +0.120]	0 (0)
OpenRCA	All	Alert-count vs RCD	1	+0.089 [+0.050, +0.127]	0 (0)
OpenRCA	All	Alert-count vs RCD	2	+0.091 [+0.055, +0.127]	0 (0)
OpenRCA	All	BARO vs RCD	0	+0.025 [+0.001, +0.049]	0 (0)
OpenRCA	All	BARO vs RCD	1	+0.027 [-0.004, +0.057]	1 (0)
OpenRCA	All	BARO vs RCD	2	+0.029 [+0.002, +0.057]	1 (0)
OpenRCA	All	CD-1min vs RCD	0	+0.068 [+0.029, +0.108]	1 (0)
OpenRCA	All	CD-1min vs RCD	1	+0.070 [+0.029, +0.113]	1 (0)
OpenRCA	All	CD-1min vs RCD	2	+0.073 [+0.032, +0.115]	1 (0)
OpenRCA	All	ϵ -Diagnosis vs RCD	0	-0.061 [-0.092, -0.032]	0 (0)
OpenRCA	All	ϵ -Diagnosis vs RCD	1	-0.059 [-0.094, -0.026]	1 (0)
OpenRCA	All	ϵ -Diagnosis vs RCD	2	-0.056 [-0.088, -0.027]	0 (0)
OpenRCA	All	max- Z vs RCD	0	+0.078 [+0.040, +0.114]	0 (0)
OpenRCA	All	max- Z vs RCD	1	+0.080 [+0.040, +0.119]	0 (0)
OpenRCA	All	max- Z vs RCD	2	+0.083 [+0.044, +0.121]	0 (0)

Per-subsystem differences

Accuracy of each published method on each subsystem, and the per-subsystem difference, for published-method pairs. Source: [paper/data/rq1/pair_subsystem_deltas.csv](#).

Table 10: Per-subsystem accuracy differences for published-method pairs (seed-averaged). Δ is accuracy of A minus accuracy of B, with its 95% CI. A row marked * has a Δ whose sign is opposite to that pair's equal-subsystem-weight pooled Δ .

Family	Pair	Subsystem	n	Acc. A	Acc. B	Δ [CI]
RCAEval RE1	BARO vs CausalRCA	RE1-OB	125	0.736	0.616	+0.120 [+0.043, +0.195]
RCAEval RE1	BARO vs CausalRCA	RE1-SS	125	0.848	0.176	+0.672 [+0.592, +0.752]
RCAEval RE1	BARO vs CausalRCA	RE1-TT	125	0.560	0.136	+0.424 [+0.331, +0.515]
RCAEval RE1	BARO vs CIRCA	RE1-OB	125	0.736	0.672	+0.064 [-0.032, +0.160]
RCAEval RE1	BARO vs CIRCA	RE1-SS	125	0.848	0.704	+0.144 [+0.064, +0.232]
RCAEval RE1	BARO vs CIRCA	RE1-TT	125	0.560	0.328	+0.232 [+0.136, +0.328]
RCAEval RE1	BARO vs ϵ -Diagnosis	RE1-OB	125	0.736	0.032	+0.704 [+0.624, +0.784]
RCAEval RE1	BARO vs ϵ -Diagnosis	RE1-SS	125	0.848	0.144	+0.704 [+0.616, +0.784]
RCAEval RE1	BARO vs ϵ -Diagnosis	RE1-TT	125	0.560	0.000	+0.560 [+0.472, +0.640]
RCAEval RE1	BARO vs RCD	RE1-OB	125	0.736	0.541	+0.195 [+0.099, +0.288]
RCAEval RE1	BARO vs RCD	RE1-SS	125	0.848	0.389	+0.459 [+0.355, +0.560]
RCAEval RE1	BARO vs RCD	RE1-TT	125	0.560	0.088	+0.472 [+0.381, +0.560]
RCAEval RE1	CausalRCA vs CIRCA	RE1-OB	125	0.616	0.672	-0.056 [-0.144, +0.035]
RCAEval RE1	CausalRCA vs CIRCA	RE1-SS	125	0.176	0.704	-0.528 [-0.613, -0.440]
RCAEval RE1	CausalRCA vs CIRCA	RE1-TT	125	0.136	0.328	-0.192 [-0.280, -0.104]
RCAEval RE1	CausalRCA vs ϵ -Diagnosis	RE1-OB	125	0.616	0.032	+0.584 [+0.512, +0.659]
RCAEval RE1	CausalRCA vs ϵ -Diagnosis	RE1-SS	125	0.176	0.144	+0.032 [-0.051, +0.117]

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Table 10: Per-subsystem accuracy differences for published-method pairs (seed-averaged). Δ is accuracy of A minus accuracy of B, with its 95% CI. A row marked * has a Δ whose sign is opposite to that pair's equal-subsystem-weight pooled Δ .

Family	Pair	Subsystem	n	Acc. A	Acc. B	Δ [CI]
RCAEval RE1	CausalRCA vs ϵ -Diagnosis	RE1-TT	125	0.136	0.000	+0.136 [+0.096, +0.179]
RCAEval RE1	CausalRCA vs RCD	RE1-OB	125	0.616	0.541	+0.075 [-0.019, +0.165]*
RCAEval RE1	CausalRCA vs RCD	RE1-SS	125	0.176	0.389	-0.213 [-0.299, -0.131]
RCAEval RE1	CausalRCA vs RCD	RE1-TT	125	0.136	0.088	+0.048 [-0.019, +0.112]*
RCAEval RE1	CIRCA vs ϵ -Diagnosis	RE1-OB	125	0.672	0.032	+0.640 [+0.552, +0.720]
RCAEval RE1	CIRCA vs ϵ -Diagnosis	RE1-SS	125	0.704	0.144	+0.560 [+0.464, +0.648]
RCAEval RE1	CIRCA vs ϵ -Diagnosis	RE1-TT	125	0.328	0.000	+0.328 [+0.248, +0.408]
RCAEval RE1	CIRCA vs RCD	RE1-OB	125	0.672	0.541	+0.131 [+0.021, +0.237]
RCAEval RE1	CIRCA vs RCD	RE1-SS	125	0.704	0.389	+0.315 [+0.195, +0.432]
RCAEval RE1	CIRCA vs RCD	RE1-TT	125	0.328	0.088	+0.240 [+0.149, +0.328]
RCAEval RE1	ϵ -Diagnosis vs RCD	RE1-OB	125	0.032	0.541	-0.509 [-0.595, -0.424]
RCAEval RE1	ϵ -Diagnosis vs RCD	RE1-SS	125	0.144	0.389	-0.245 [-0.352, -0.141]
RCAEval RE1	ϵ -Diagnosis vs RCD	RE1-TT	125	0.000	0.088	-0.088 [-0.136, -0.048]
RCAEval RE2	BARO vs CausalRCA	RE2-OB	90	0.144	0.207	-0.063 [-0.159, +0.041]*
RCAEval RE2	BARO vs CausalRCA	RE2-SS	90	0.067	0.148	-0.081 [-0.163, -0.004]*
RCAEval RE2	BARO vs CausalRCA	RE2-TT	90	0.667	0.207	+0.459 [+0.341, +0.574]
RCAEval RE2	BARO vs CIRCA	RE2-OB	90	0.144	0.667	-0.522 [-0.644, -0.400]
RCAEval RE2	BARO vs CIRCA	RE2-SS	90	0.067	0.778	-0.711 [-0.800, -0.611]
RCAEval RE2	BARO vs CIRCA	RE2-TT	90	0.667	0.544	+0.122 [+0.011, +0.233]*
RCAEval RE2	BARO vs ϵ -Diagnosis	RE2-OB	90	0.144	0.033	+0.111 [+0.022, +0.200]
RCAEval RE2	BARO vs ϵ -Diagnosis	RE2-SS	90	0.067	0.156	-0.089 [-0.189, +0.000]*
RCAEval RE2	BARO vs ϵ -Diagnosis	RE2-TT	90	0.667	0.000	+0.667 [+0.567, +0.767]
RCAEval RE2	BARO vs MicroRank	RE2-OB	90	0.144	0.000	+0.144 [+0.078, +0.222]
RCAEval RE2	BARO vs MicroRank	RE2-TT	90	0.667	0.122	+0.544 [+0.411, +0.667]
RCAEval RE2	BARO vs BARO (multi-source)	RE2-OB	90	0.144	0.167	-0.022 [-0.067, +0.022]
RCAEval RE2	BARO vs BARO (multi-source)	RE2-SS	90	0.067	0.067	+0.000 [-0.033, +0.033]
RCAEval RE2	BARO vs BARO (multi-source)	RE2-TT	90	0.667	0.689	-0.022 [-0.078, +0.033]
RCAEval RE2	BARO vs SimpleRCA	RE2-OB	90	0.144	0.467	-0.322 [-0.433, -0.211]
RCAEval RE2	BARO vs SimpleRCA	RE2-SS	90	0.067	0.244	-0.178 [-0.278, -0.089]
RCAEval RE2	BARO vs SimpleRCA	RE2-TT	90	0.667	0.800	-0.133 [-0.222, -0.044]
RCAEval RE2	BARO vs TraceRCA	RE2-OB	90	0.144	0.100	+0.044 [-0.056, +0.144]
RCAEval RE2	BARO vs TraceRCA	RE2-TT	90	0.667	0.533	+0.133 [-0.011, +0.267]
RCAEval RE2	CausalRCA vs CIRCA	RE2-OB	90	0.207	0.667	-0.459 [-0.567, -0.356]
RCAEval RE2	CausalRCA vs CIRCA	RE2-SS	90	0.148	0.778	-0.630 [-0.722, -0.530]
RCAEval RE2	CausalRCA vs CIRCA	RE2-TT	90	0.207	0.544	-0.337 [-0.456, -0.219]
RCAEval RE2	CausalRCA vs ϵ -Diagnosis	RE2-OB	90	0.207	0.033	+0.174 [+0.089, +0.259]
RCAEval RE2	CausalRCA vs ϵ -Diagnosis	RE2-SS	90	0.148	0.156	-0.007 [-0.104, +0.089]*
RCAEval RE2	CausalRCA vs ϵ -Diagnosis	RE2-TT	90	0.207	0.000	+0.207 [+0.152, +0.267]
RCAEval RE2	CausalRCA vs MicroRank	RE2-OB	90	0.207	0.000	+0.207 [+0.137, +0.278]
RCAEval RE2	CausalRCA vs MicroRank	RE2-TT	90	0.207	0.122	+0.085 [-0.007, +0.174]
RCAEval RE2	CausalRCA vs BARO (multi-source)	RE2-OB	90	0.207	0.167	+0.041 [-0.067, +0.141]*
RCAEval RE2	CausalRCA vs BARO (multi-source)	RE2-SS	90	0.148	0.067	+0.081 [+0.004, +0.163]*
RCAEval RE2	CausalRCA vs BARO (multi-source)	RE2-TT	90	0.207	0.689	-0.481 [-0.596, -0.363]
RCAEval RE2	CausalRCA vs SimpleRCA	RE2-OB	90	0.207	0.467	-0.259 [-0.374, -0.148]
RCAEval RE2	CausalRCA vs SimpleRCA	RE2-SS	90	0.148	0.244	-0.096 [-0.204, +0.007]
RCAEval RE2	CausalRCA vs SimpleRCA	RE2-TT	90	0.207	0.800	-0.593 [-0.681, -0.496]
RCAEval RE2	CausalRCA vs TraceRCA	RE2-OB	90	0.207	0.100	+0.107 [+0.011, +0.204]*
RCAEval RE2	CausalRCA vs TraceRCA	RE2-TT	90	0.207	0.533	-0.326 [-0.437, -0.215]
RCAEval RE2	CIRCA vs ϵ -Diagnosis	RE2-OB	90	0.667	0.033	+0.633 [+0.533, +0.744]
RCAEval RE2	CIRCA vs ϵ -Diagnosis	RE2-SS	90	0.778	0.156	+0.622 [+0.500, +0.733]
RCAEval RE2	CIRCA vs ϵ -Diagnosis	RE2-TT	90	0.544	0.000	+0.544 [+0.444, +0.644]
RCAEval RE2	CIRCA vs MicroRank	RE2-OB	90	0.667	0.000	+0.667 [+0.567, +0.767]
RCAEval RE2	CIRCA vs MicroRank	RE2-TT	90	0.544	0.122	+0.422 [+0.300, +0.544]
RCAEval RE2	CIRCA vs BARO (multi-source)	RE2-OB	90	0.667	0.167	+0.500 [+0.378, +0.622]
RCAEval RE2	CIRCA vs BARO (multi-source)	RE2-SS	90	0.778	0.067	+0.711 [+0.622, +0.800]
RCAEval RE2	CIRCA vs BARO (multi-source)	RE2-TT	90	0.544	0.689	-0.144 [-0.267, -0.022]*
RCAEval RE2	CIRCA vs SimpleRCA	RE2-OB	90	0.667	0.467	+0.200 [+0.078, +0.322]
RCAEval RE2	CIRCA vs SimpleRCA	RE2-SS	90	0.778	0.244	+0.533 [+0.422, +0.644]
RCAEval RE2	CIRCA vs SimpleRCA	RE2-TT	90	0.544	0.800	-0.256 [-0.378, -0.133]*
RCAEval RE2	CIRCA vs TraceRCA	RE2-OB	90	0.667	0.100	+0.567 [+0.456, +0.678]
RCAEval RE2	CIRCA vs TraceRCA	RE2-TT	90	0.544	0.533	+0.011 [-0.144, +0.167]
RCAEval RE2	ϵ -Diagnosis vs MicroRank	RE2-OB	90	0.033	0.000	+0.033 [+0.000, +0.078]*
RCAEval RE2	ϵ -Diagnosis vs MicroRank	RE2-TT	90	0.000	0.122	-0.122 [-0.189, -0.056]
RCAEval RE2	ϵ -Diagnosis vs BARO (multi-source)	RE2-OB	90	0.033	0.167	-0.133 [-0.222, -0.044]
RCAEval RE2	ϵ -Diagnosis vs BARO (multi-source)	RE2-SS	90	0.156	0.067	+0.089 [-0.000, +0.189]*
RCAEval RE2	ϵ -Diagnosis vs BARO (multi-source)	RE2-TT	90	0.000	0.689	-0.689 [-0.789, -0.589]
RCAEval RE2	ϵ -Diagnosis vs SimpleRCA	RE2-OB	90	0.033	0.467	-0.433 [-0.544, -0.322]
RCAEval RE2	ϵ -Diagnosis vs SimpleRCA	RE2-SS	90	0.156	0.244	-0.089 [-0.211, +0.033]
RCAEval RE2	ϵ -Diagnosis vs SimpleRCA	RE2-TT	90	0.000	0.800	-0.800 [-0.878, -0.711]
RCAEval RE2	ϵ -Diagnosis vs TraceRCA	RE2-OB	90	0.033	0.100	-0.067 [-0.144, +0.011]
RCAEval RE2	ϵ -Diagnosis vs TraceRCA	RE2-TT	90	0.000	0.533	-0.533 [-0.633, -0.433]
RCAEval RE2	MicroRank vs BARO (multi-source)	RE2-OB	90	0.000	0.167	-0.167 [-0.244, -0.089]
RCAEval RE2	MicroRank vs BARO (multi-source)	RE2-TT	90	0.122	0.689	-0.567 [-0.689, -0.433]
RCAEval RE2	MicroRank vs SimpleRCA	RE2-OB	90	0.000	0.467	-0.467 [-0.567, -0.367]
RCAEval RE2	MicroRank vs SimpleRCA	RE2-TT	90	0.122	0.800	-0.678 [-0.789, -0.544]
RCAEval RE2	MicroRank vs TraceRCA	RE2-OB	90	0.000	0.100	-0.100 [-0.167, -0.044]
RCAEval RE2	MicroRank vs TraceRCA	RE2-TT	90	0.122	0.533	-0.411 [-0.522, -0.300]
RCAEval RE2	BARO (multi-source) vs SimpleRCA	RE2-OB	90	0.167	0.467	-0.300 [-0.411, -0.178]

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Table 10: Per-subsystem accuracy differences for published-method pairs (seed-averaged). Δ is accuracy of A minus accuracy of B, with its 95% CI. A row marked * has a Δ whose sign is opposite to that pair’s equal-subsystem-weight pooled Δ .

Family	Pair	Subsystem	n	Acc. A	Acc. B	Δ [CI]
RCAEval RE2	BARO (multi-source) vs SimpleRCA	RE2-SS	90	0.067	0.244	-0.178 [-0.278, -0.078]
RCAEval RE2	BARO (multi-source) vs SimpleRCA	RE2-TT	90	0.689	0.800	-0.111 [-0.200, -0.033]
RCAEval RE2	BARO (multi-source) vs TraceRCA	RE2-OB	90	0.167	0.100	+0.067 [-0.044, +0.178]
RCAEval RE2	BARO (multi-source) vs TraceRCA	RE2-TT	90	0.689	0.533	+0.156 [+0.011, +0.300]
RCAEval RE2	SimpleRCA vs TraceRCA	RE2-OB	90	0.467	0.100	+0.367 [+0.244, +0.489]
RCAEval RE2	SimpleRCA vs TraceRCA	RE2-TT	90	0.800	0.533	+0.267 [+0.122, +0.400]
OpenRCA	BARO vs ϵ -Diagnosis	Bank	136	0.160	0.036	+0.125 [+0.067, +0.184]
OpenRCA	BARO vs ϵ -Diagnosis	Market-1	70	0.057	0.038	+0.019 [-0.020, +0.059]
OpenRCA	BARO vs ϵ -Diagnosis	Market-2	78	0.106	0.044	+0.062 [+0.005, +0.122]
OpenRCA	BARO vs ϵ -Diagnosis	Telecom	51	0.183	0.046	+0.137 [+0.052, +0.232]
OpenRCA	BARO vs RCD	Bank	136	0.160	0.089	+0.071 [+0.018, +0.127]
OpenRCA	BARO vs RCD	Market-1	70	0.057	0.051	+0.006 [-0.035, +0.049]
OpenRCA	BARO vs RCD	Market-2	78	0.106	0.066	+0.040 [-0.013, +0.093]
OpenRCA	BARO vs RCD	Telecom	51	0.183	0.193	-0.010 [-0.058, +0.035]*
OpenRCA	ϵ -Diagnosis vs RCD	Bank	136	0.036	0.089	-0.054 [-0.091, -0.018]
OpenRCA	ϵ -Diagnosis vs RCD	Market-1	70	0.038	0.051	-0.013 [-0.054, +0.027]
OpenRCA	ϵ -Diagnosis vs RCD	Market-2	78	0.044	0.066	-0.023 [-0.076, +0.025]
OpenRCA	ϵ -Diagnosis vs RCD	Telecom	51	0.046	0.193	-0.147 [-0.241, -0.062]
PetShop	BARO vs CIRCA	High	26	0.154	0.269	-0.115 [-0.269, +0.038]
PetShop	BARO vs CIRCA	Low	26	0.154	0.385	-0.231 [-0.423, -0.038]
PetShop	BARO vs CIRCA	Temporal-1	8	0.250	0.500	-0.250 [-0.750, +0.250]
PetShop	BARO vs CIRCA	Temporal-2	8	0.250	0.625	-0.375 [-0.875, +0.125]
PetShop	BARO vs Counterfactual	High	26	0.154	0.308	-0.154 [-0.346, +0.038]
PetShop	BARO vs Counterfactual	Low	26	0.154	0.154	+0.000 [-0.231, +0.192]
PetShop	BARO vs Counterfactual	Temporal-1	8	0.250	0.125	+0.125 [-0.250, +0.500]*
PetShop	BARO vs Counterfactual	Temporal-2	8	0.250	0.250	+0.000 [-0.375, +0.375]
PetShop	BARO vs ϵ -Diagnosis	High	26	0.154	0.000	+0.154 [+0.038, +0.308]
PetShop	BARO vs ϵ -Diagnosis	Low	26	0.154	0.000	+0.154 [+0.038, +0.308]
PetShop	BARO vs ϵ -Diagnosis	Temporal-1	8	0.250	0.000	+0.250 [+0.000, +0.625]
PetShop	BARO vs ϵ -Diagnosis	Temporal-2	8	0.250	0.125	+0.125 [-0.250, +0.500]
PetShop	BARO vs RCD	High	26	0.154	0.000	+0.154 [+0.038, +0.308]*
PetShop	BARO vs RCD	Low	26	0.154	0.346	-0.192 [-0.423, +0.038]
PetShop	BARO vs RCD	Temporal-1	8	0.250	0.625	-0.375 [-0.875, +0.125]
PetShop	BARO vs RCD	Temporal-2	8	0.250	0.375	-0.125 [-0.500, +0.250]
PetShop	CIRCA vs Counterfactual	High	26	0.269	0.308	-0.038 [-0.269, +0.192]*
PetShop	CIRCA vs Counterfactual	Low	26	0.385	0.154	+0.231 [-0.038, +0.500]
PetShop	CIRCA vs Counterfactual	Temporal-1	8	0.500	0.125	+0.375 [+0.125, +0.750]
PetShop	CIRCA vs Counterfactual	Temporal-2	8	0.625	0.250	+0.375 [-0.125, +0.875]
PetShop	CIRCA vs ϵ -Diagnosis	High	26	0.269	0.000	+0.269 [+0.115, +0.462]
PetShop	CIRCA vs ϵ -Diagnosis	Low	26	0.385	0.000	+0.385 [+0.192, +0.577]
PetShop	CIRCA vs ϵ -Diagnosis	Temporal-1	8	0.500	0.000	+0.500 [+0.125, +0.875]
PetShop	CIRCA vs ϵ -Diagnosis	Temporal-2	8	0.625	0.125	+0.500 [+0.125, +0.875]
PetShop	CIRCA vs RCD	High	26	0.269	0.000	+0.269 [+0.115, +0.462]
PetShop	CIRCA vs RCD	Low	26	0.385	0.346	+0.038 [-0.231, +0.308]
PetShop	CIRCA vs RCD	Temporal-1	8	0.500	0.625	-0.125 [-0.625, +0.375]*
PetShop	CIRCA vs RCD	Temporal-2	8	0.625	0.375	+0.250 [-0.250, +0.750]
PetShop	Counterfactual vs ϵ -Diagnosis	High	26	0.308	0.000	+0.308 [+0.154, +0.500]
PetShop	Counterfactual vs ϵ -Diagnosis	Low	26	0.154	0.000	+0.154 [+0.038, +0.308]
PetShop	Counterfactual vs ϵ -Diagnosis	Temporal-1	8	0.125	0.000	+0.125 [+0.000, +0.375]
PetShop	Counterfactual vs ϵ -Diagnosis	Temporal-2	8	0.250	0.125	+0.125 [+0.000, +0.375]
PetShop	Counterfactual vs RCD	High	26	0.308	0.000	+0.308 [+0.154, +0.500]*
PetShop	Counterfactual vs RCD	Low	26	0.154	0.346	-0.192 [-0.423, +0.038]
PetShop	Counterfactual vs RCD	Temporal-1	8	0.125	0.625	-0.500 [-0.875, +0.000]
PetShop	Counterfactual vs RCD	Temporal-2	8	0.250	0.375	-0.125 [-0.500, +0.250]
PetShop	ϵ -Diagnosis vs RCD	High	26	0.000	0.000	+0.000 [+0.000, +0.000]
PetShop	ϵ -Diagnosis vs RCD	Low	26	0.000	0.346	-0.346 [-0.538, -0.154]
PetShop	ϵ -Diagnosis vs RCD	Temporal-1	8	0.000	0.625	-0.625 [-0.875, -0.250]
PetShop	ϵ -Diagnosis vs RCD	Temporal-2	8	0.125	0.375	-0.250 [-0.750, +0.250]

Small-sample sensitivity

How many pairs still reverse when subsystems below a minimum case count are set aside. Source: paper/data/rq1/small_sample.csv.

Table 11: Small-sample sensitivity. For each minimum case count $n_{\min}$, family, and layer, the number of pairs, the number of pairs with at least one subsystem reversal, and the number of pairs with at least one CI-supported reversal.

$n_{\min}$	Family	Layer	Pairs	Pairs with ≥ 1 reversal	With CI-supported reversal
0	RCAEval RE1	Published	10	1	0
0	RCAEval RE1	All	28	8	4
0	OpenRCA	Published	3	1	0

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Table 11: Small-sample sensitivity. For each minimum case count $n_{\min}$, family, and layer, the number of pairs, the number of pairs with at least one subsystem reversal, and the number of pairs with at least one CI-supported reversal.

$n_{\min}$	Family	Layer	Pairs	Pairs with ≥ 1 reversal	With CI-supported reversal
0	OpenRCA	All	15	7	0
0	PetShop	Published	10	5	2
0	PetShop	All	55	12	4
25	RCAEval RE1	Published	10	1	0
25	RCAEval RE1	All	28	8	4
25	OpenRCA	Published	3	1	0
25	OpenRCA	All	15	7	0
25	PetShop	Published	10	3	1
25	PetShop	All	55	7	1
50	RCAEval RE1	Published	10	1	0
50	RCAEval RE1	All	28	8	4
50	OpenRCA	Published	3	1	0
50	OpenRCA	All	15	7	0

Coverage and accuracy

Valid-output coverage and accuracy under both denominators, for every subsystem, method, and seed. Source: <paper/data/rq2/coverage.csv>.

Table 12: Output coverage and accuracy for every subsystem, method, and seed. Valid is the number of cases with a usable ranking. Accuracy (all cases) uses every case as the denominator (an unusable output counts as a miss); accuracy (valid only) uses only cases with a usable ranking. A seed of – means the method was not re-run across seeds.

Subsystem	Method	Seed	n	Valid	Coverage	Acc. (all cases)	Acc. (valid only)
RE1-OB	Alert-count	–	125	125	1.000	0.392	0.392
RE1-OB	BARO	–	125	125	1.000	0.736	0.736
RE1-OB	CausalRCA	0	125	125	1.000	0.640	0.640
RE1-OB	CausalRCA	1	125	125	1.000	0.568	0.568
RE1-OB	CausalRCA	2	125	125	1.000	0.640	0.640
RE1-OB	CD-1min	–	125	125	1.000	0.728	0.728
RE1-OB	CIRCA	–	125	125	1.000	0.672	0.672
RE1-OB	ϵ -Diagnosis	–	125	125	1.000	0.032	0.032
RE1-OB	max- Z	–	125	125	1.000	0.504	0.504
RE1-OB	RCD	0	125	125	1.000	0.528	0.528
RE1-OB	RCD	1	125	125	1.000	0.568	0.568
RE1-OB	RCD	2	125	125	1.000	0.528	0.528
RE1-SS	Alert-count	–	125	125	1.000	0.376	0.376
RE1-SS	BARO	–	125	125	1.000	0.848	0.848
RE1-SS	CausalRCA	0	125	125	1.000	0.256	0.256
RE1-SS	CausalRCA	1	125	125	1.000	0.152	0.152
RE1-SS	CausalRCA	2	125	125	1.000	0.120	0.120
RE1-SS	CD-1min	–	125	125	1.000	0.928	0.928
RE1-SS	CIRCA	–	125	125	1.000	0.704	0.704
RE1-SS	ϵ -Diagnosis	–	125	125	1.000	0.144	0.144
RE1-SS	max- Z	–	125	125	1.000	0.744	0.744
RE1-SS	RCD	0	125	125	1.000	0.376	0.376
RE1-SS	RCD	1	125	125	1.000	0.384	0.384
RE1-SS	RCD	2	125	125	1.000	0.408	0.408
RE1-TT	Alert-count	–	125	125	1.000	0.048	0.048
RE1-TT	BARO	–	125	125	1.000	0.560	0.560
RE1-TT	CausalRCA	0	125	125	1.000	0.160	0.160
RE1-TT	CausalRCA	1	125	125	1.000	0.208	0.208
RE1-TT	CausalRCA	2	125	125	1.000	0.040	0.040
RE1-TT	CD-1min	–	125	125	1.000	0.280	0.280
RE1-TT	CIRCA	–	125	125	1.000	0.328	0.328
RE1-TT	ϵ -Diagnosis	–	125	125	1.000	0.000	0.000
RE1-TT	max- Z	–	125	125	1.000	0.424	0.424
RE1-TT	RCD	0	125	125	1.000	0.072	0.072
RE1-TT	RCD	1	125	125	1.000	0.080	0.080
RE1-TT	RCD	2	125	125	1.000	0.112	0.112
RE2-OB	Alert-count	–	90	90	1.000	0.211	0.211
RE2-OB	BARO	–	90	90	1.000	0.144	0.144
RE2-OB	CausalRCA	0	90	90	1.000	0.278	0.278
RE2-OB	CausalRCA	1	90	90	1.000	0.156	0.156
RE2-OB	CausalRCA	2	90	90	1.000	0.189	0.189
RE2-OB	CD-1min	–	90	90	1.000	0.733	0.733
RE2-OB	CIRCA	–	90	90	1.000	0.667	0.667
RE2-OB	ϵ -Diagnosis	–	90	90	1.000	0.033	0.033
RE2-OB	max- Z	–	90	90	1.000	0.778	0.778
RE2-OB	MicroRank	–	90	90	1.000	0.000	0.000
RE2-OB	BARO (multi-source)	–	90	90	1.000	0.167	0.167

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Table 12: Output coverage and accuracy for every subsystem, method, and seed. Valid is the number of cases with a usable ranking. Accuracy (all cases) uses every case as the denominator (an unusable output counts as a miss); accuracy (valid only) uses only cases with a usable ranking. A seed of – means the method was not re-run across seeds.

Subsystem	Method	Seed	n	Valid	Coverage	Acc. (all cases)	Acc. (valid only)
RE2-OB	SimpleRCA	–	90	90	1.000	0.467	0.467
RE2-OB	TraceRCA	–	90	90	1.000	0.100	0.100
RE2-SS	Alert-count	–	90	90	1.000	0.389	0.389
RE2-SS	BARO	–	90	90	1.000	0.067	0.067
RE2-SS	CausalRCA	0	90	90	1.000	0.144	0.144
RE2-SS	CausalRCA	1	90	90	1.000	0.178	0.178
RE2-SS	CausalRCA	2	90	90	1.000	0.122	0.122
RE2-SS	CD-1min	–	90	90	1.000	0.944	0.944
RE2-SS	CIRCA	–	90	90	1.000	0.778	0.778
RE2-SS	ϵ -Diagnosis	–	90	90	1.000	0.156	0.156
RE2-SS	max- Z	–	90	90	1.000	0.844	0.844
RE2-SS	BARO (multi-source)	–	90	90	1.000	0.067	0.067
RE2-SS	SimpleRCA	–	90	90	1.000	0.244	0.244
RE2-TT	Alert-count	–	90	90	1.000	0.211	0.211
RE2-TT	BARO	–	90	90	1.000	0.667	0.667
RE2-TT	CausalRCA	0	90	90	1.000	0.267	0.267
RE2-TT	CausalRCA	1	90	47	0.522	0.133	0.255
RE2-TT	CausalRCA	2	90	90	1.000	0.222	0.222
RE2-TT	CD-1min	–	90	90	1.000	0.789	0.789
RE2-TT	CIRCA	–	90	90	1.000	0.544	0.544
RE2-TT	ϵ -Diagnosis	–	90	90	1.000	0.000	0.000
RE2-TT	max- Z	–	90	90	1.000	0.411	0.411
RE2-TT	MicroRank	–	90	73	0.811	0.122	0.151
RE2-TT	BARO (multi-source)	–	90	89	0.989	0.689	0.697
RE2-TT	SimpleRCA	–	90	90	1.000	0.800	0.800
RE2-TT	TraceRCA	–	90	73	0.811	0.533	0.658
Bank	Alert-count	–	136	136	1.000	0.172	0.172
Bank	BARO	–	136	136	1.000	0.160	0.160
Bank	CD-1min	–	136	132	0.971	0.165	0.170
Bank	ϵ -Diagnosis	–	136	136	1.000	0.036	0.036
Bank	max- Z	–	136	136	1.000	0.127	0.127
Bank	RCD	0	136	136	1.000	0.094	0.094
Bank	RCD	1	136	136	1.000	0.087	0.087
Bank	RCD	2	136	136	1.000	0.086	0.086
Market-1	Alert-count	–	70	70	1.000	0.144	0.144
Market-1	BARO	–	70	70	1.000	0.057	0.057
Market-1	CD-1min	–	70	70	1.000	0.161	0.161
Market-1	ϵ -Diagnosis	–	70	70	1.000	0.038	0.038
Market-1	max- Z	–	70	70	1.000	0.161	0.161
Market-1	RCD	0	70	63	0.900	0.057	0.063
Market-1	RCD	1	70	62	0.886	0.036	0.040
Market-1	RCD	2	70	63	0.900	0.060	0.066
Market-2	Alert-count	–	78	78	1.000	0.197	0.197
Market-2	BARO	–	78	78	1.000	0.106	0.106
Market-2	CD-1min	–	78	78	1.000	0.192	0.192
Market-2	ϵ -Diagnosis	–	78	78	1.000	0.044	0.044
Market-2	max- Z	–	78	78	1.000	0.200	0.200
Market-2	RCD	0	78	74	0.949	0.074	0.078
Market-2	RCD	1	78	73	0.936	0.058	0.062
Market-2	RCD	2	78	74	0.949	0.067	0.071
Telecom	Alert-count	–	51	51	1.000	0.242	0.242
Telecom	BARO	–	51	51	1.000	0.183	0.183
Telecom	CD-1min	–	51	47	0.922	0.163	0.177
Telecom	ϵ -Diagnosis	–	51	19	0.373	0.046	0.123
Telecom	max- Z	–	51	51	1.000	0.232	0.232
Telecom	RCD	0	51	50	0.980	0.183	0.186
Telecom	RCD	1	51	50	0.980	0.219	0.223
Telecom	RCD	2	51	49	0.961	0.176	0.183
High	Alert-count	–	26	26	1.000	0.038	0.038
High	BARO	–	26	26	1.000	0.154	0.154
High	Traversal	–	26	24	0.923	0.385	0.417
High	CD-1min	–	26	26	1.000	0.000	0.000
High	CIRCA	–	26	26	1.000	0.269	0.269
High	Counterfactual	–	26	26	1.000	0.308	0.308
High	ϵ -Diagnosis	–	26	16	0.615	0.000	0.000
High	max- Z	–	26	26	1.000	0.077	0.077
High	Random walk	–	26	24	0.923	0.000	0.000
High	Ranked correlation	–	26	24	0.923	0.731	0.792
High	RCD	–	26	26	1.000	0.000	0.000
Low	Alert-count	–	26	26	1.000	0.000	0.000
Low	BARO	–	26	26	1.000	0.154	0.154
Low	Traversal	–	26	23	0.885	0.577	0.652
Low	CD-1min	–	26	26	1.000	0.038	0.038
Low	CIRCA	–	26	26	1.000	0.385	0.385
Low	Counterfactual	–	26	26	1.000	0.154	0.154
Low	ϵ -Diagnosis	–	26	17	0.654	0.000	0.000
Low	max- Z	–	26	26	1.000	0.192	0.192
Low	Random walk	–	26	23	0.885	0.000	0.000

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Table 12: Output coverage and accuracy for every subsystem, method, and seed. Valid is the number of cases with a usable ranking. Accuracy (all cases) uses every case as the denominator (an unusable output counts as a miss); accuracy (valid only) uses only cases with a usable ranking. A seed of – means the method was not re-run across seeds.

Subsystem	Method	Seed	n	Valid	Coverage	Acc. (all cases)	Acc. (valid only)
Low	Ranked correlation	–	26	23	0.885	0.577	0.652
Low	RCD	–	26	26	1.000	0.346	0.346
Temporal-1	Alert-count	–	8	8	1.000	0.000	0.000
Temporal-1	BARO	–	8	8	1.000	0.250	0.250
Temporal-1	Traversal	–	8	8	1.000	0.625	0.625
Temporal-1	CD-1min	–	8	8	1.000	0.125	0.125
Temporal-1	CIRCA	–	8	8	1.000	0.500	0.500
Temporal-1	Counterfactual	–	8	8	1.000	0.125	0.125
Temporal-1	ϵ -Diagnosis	–	8	6	0.750	0.000	0.000
Temporal-1	max- Z	–	8	8	1.000	0.375	0.375
Temporal-1	Random walk	–	8	8	1.000	0.125	0.125
Temporal-1	Ranked correlation	–	8	8	1.000	0.750	0.750
Temporal-1	RCD	–	8	8	1.000	0.625	0.625
Temporal-2	Alert-count	–	8	8	1.000	0.250	0.250
Temporal-2	BARO	–	8	8	1.000	0.250	0.250
Temporal-2	Traversal	–	8	8	1.000	0.875	0.875
Temporal-2	CD-1min	–	8	8	1.000	0.125	0.125
Temporal-2	CIRCA	–	8	8	1.000	0.625	0.625
Temporal-2	Counterfactual	–	8	8	1.000	0.250	0.250
Temporal-2	ϵ -Diagnosis	–	8	8	1.000	0.125	0.125
Temporal-2	max- Z	–	8	8	1.000	0.375	0.375
Temporal-2	Random walk	–	8	8	1.000	0.000	0.000
Temporal-2	Ranked correlation	–	8	8	1.000	0.500	0.500
Temporal-2	RCD	–	8	8	1.000	0.375	0.375

Input representation

Accuracy on the reduced input file and on the raw export, with the file names. Source: `paper/data/rq2/input_accuracy.csv`.

Table 13: Accuracy under the reduced (simple) input released for analysis and under the raw export. The Files column names the reduced file and the raw file.

Subsystem	Method	Acc. reduced input	Acc. raw input	Files (reduced / raw)
RE1-SS	Alert-count	0.376	0.432	simple_data.csv / data.csv
RE1-SS	BARO	0.848	0.200	simple_data.csv / data.csv
RE1-SS	CD-1min	0.928	0.736	simple_data.csv / data.csv
RE1-SS	CIRCA	0.704	0.200	simple_data.csv / data.csv
RE1-SS	ϵ -Diagnosis	0.144	0.168	simple_data.csv / data.csv
RE1-SS	max- Z	0.744	0.608	simple_data.csv / data.csv
RE1-SS	RCD	0.389	0.296	simple_data.csv / data.csv
RE1-TT	Alert-count	0.048	0.192	simple_data.csv / data.csv
RE1-TT	BARO	0.560	0.120	simple_data.csv / data.csv
RE1-TT	CD-1min	0.280	0.456	simple_data.csv / data.csv
RE1-TT	CIRCA	0.328	0.000	simple_data.csv / data.csv
RE1-TT	ϵ -Diagnosis	0.000	0.000	simple_data.csv / data.csv
RE1-TT	max- Z	0.424	0.448	simple_data.csv / data.csv
RE1-TT	RCD	0.088	0.091	simple_data.csv / data.csv
RE2-OB	Alert-count	0.211	0.289	simple_metrics.csv / metrics.csv
RE2-OB	CD-1min	0.733	0.400	simple_metrics.csv / metrics.csv
RE2-OB	max- Z	0.778	0.522	simple_metrics.csv / metrics.csv
RE2-OB	BARO (multi-source)	0.167	0.233	simple_metrics.csv + logs.csv + tracets_lat,err.csv / metrics.csv + logs.csv + tracets_lat,err.csv
RE2-SS	Alert-count	0.389	0.356	simple_metrics.csv / metrics.csv
RE2-SS	CD-1min	0.944	0.533	simple_metrics.csv / metrics.csv
RE2-SS	max- Z	0.844	0.556	simple_metrics.csv / metrics.csv
RE2-SS	BARO (multi-source)	0.067	0.267	simple_metrics.csv + logs.csv + tracets_lat,err.csv / metrics.csv + logs.csv + tracets_lat,err.csv
RE2-TT	Alert-count	0.211	0.278	simple_metrics.csv / metrics.csv
RE2-TT	CD-1min	0.789	0.244	simple_metrics.csv / metrics.csv
RE2-TT	max- Z	0.411	0.222	simple_metrics.csv / metrics.csv
RE2-TT	BARO (multi-source)	0.689	0.144	simple_metrics.csv + logs.csv + tracets_lat,err.csv / metrics.csv + logs.csv + tracets_lat,err.csv

OpenRCA scoring variants

Partial-credit, exact-match, and at-least-0.5 accuracy on each OpenRCA subsystem. Source: `paper/data/rq2/openrca_scoring_accuracy.csv`.

Table 4: Held-out subsystems: adjacent pairs of the leaderboard built on the other subsystems of a family. A decision is passed when the margin on the remaining subsystems is at least c times their spread ($c = 1$ is the line of the main text; $c = 0$ passes every decision with a direction). Wrong: the higher-ranked method is less accurate on the held-out subsystem. The last two columns pass the same number of decisions by the largest system-equal and case-weighted pooled margins.

Layer	Family	c	Decisions	Passed	Wrong	Flagged	Wrong	Margin only (equal) wrong	Margin only (case-wtd.) wrong
Published	All three	0.0	36	36	8	0	0	8	10
		0.5	36	26	6	10	2	5	7
		1.0	36	19	4	17	4	3	3
		1.5	36	10	2	26	6	2	2
		2.0	36	7	1	29	7	2	1
	RCAEval RE1	0.0	12	12	3	0	0	3	3
		0.5	12	10	1	2	2	1	1
		1.0	12	9	1	3	2	0	0
		1.5	12	5	1	7	2	0	0
		2.0	12	3	1	9	2	0	0
	OpenRCA	0.0	8	8	1	0	0	1	1
		0.5	8	7	1	1	0	1	1
		1.0	8	4	1	4	0	1	1
		1.5	8	0	0	8	1	0	0
		2.0	8	0	0	8	1	0	0
	PetShop	0.0	16	16	4	0	0	4	6
		0.5	16	9	4	7	0	2	2
		1.0	16	6	2	10	2	2	2
		1.5	16	5	1	11	3	2	2
		2.0	16	4	0	12	4	2	2
All pairs	All three	0.0	81	79	27	2	0	27	27
		0.5	81	44	13	37	14	14	15
		1.0	81	20	6	61	21	5	4
		1.5	81	9	3	72	24	2	2
		2.0	81	6	1	75	26	1	1
	RCAEval RE1	0.0	21	21	8	0	0	8	8
		0.5	21	12	4	9	4	3	3
		1.0	21	8	2	13	6	1	1
		1.5	21	5	2	16	6	1	1
		2.0	21	3	1	18	7	0	0
	OpenRCA	0.0	20	20	6	0	0	6	5
		0.5	20	10	2	10	4	1	1
		1.0	20	4	1	16	5	0	0
		1.5	20	0	0	20	6	0	0
		2.0	20	0	0	20	6	0	0
	PetShop	0.0	40	38	13	2	0	13	14
		0.5	40	22	7	18	6	6	7
		1.0	40	8	3	32	10	3	3
		1.5	40	4	1	36	12	2	1
		2.0	40	3	0	37	13	1	1

Table 14: OpenRCA scoring variants. Partial credit is the released fractional score; strict counts only a score of exactly 1; ≥ 0.5 counts a score of at least 0.5.

Subsystem	Method	Partial credit	Strict (= 1)	≥ 0.5
Bank	Alert-count	0.172	0.081	0.235
Bank	BARO	0.160	0.096	0.169
Bank	CD-1min	0.165	0.096	0.191
Bank	ϵ -Diagnosis	0.036	0.022	0.044
Bank	max- Z	0.127	0.059	0.162
Bank	RCD	0.089	0.049	0.081
Market-1	Alert-count	0.144	0.014	0.214
Market-1	BARO	0.057	0.000	0.071
Market-1	CD-1min	0.161	0.057	0.214
Market-1	ϵ -Diagnosis	0.038	0.000	0.043
Market-1	max- Z	0.161	0.029	0.229
Market-1	RCD	0.051	0.000	0.043
Market-2	Alert-count	0.197	0.090	0.269
Market-2	BARO	0.106	0.026	0.167
Market-2	CD-1min	0.192	0.115	0.231
Market-2	ϵ -Diagnosis	0.044	0.000	0.051
Market-2	max- Z	0.200	0.077	0.282
Market-2	RCD	0.066	0.026	0.051
Telecom	Alert-count	0.242	0.157	0.255
Telecom	BARO	0.183	0.118	0.196
Telecom	CD-1min	0.163	0.118	0.157

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Table 14: OpenRCA scoring variants. Partial credit is the released fractional score; strict counts only a score of exactly 1; ≥ 0.5 counts a score of at least 0.5.

Subsystem	Method	Partial credit	Strict (= 1)	≥ 0.5
Telecom	ϵ -Diagnosis	0.046	0.020	0.059
Telecom	max- Z	0.232	0.137	0.275
Telecom	RCD	0.193	0.124	0.216

Duplicated header: fixed vs as shipped

Accuracy with the RE1-OB duplicate header fixed and as shipped, and the pair contrasts whose sign or reversal status changes. Source: `paper/data/rq2/release_defect_accuracy.csv`.

Table 15: Accuracy with the released RE1-OB duplicate-header silent failure left as shipped, and with that header fixed. Affected columns restrict both accuracies to the cases whose input row is changed by the fix.

Method	Acc. fixed	Acc. as shipped	Acc. fixed (affected)	Acc. as shipped (affected)	n affected
CIRCA	0.672	0.432	0.820	0.220	50
ϵ -Diagnosis	0.032	0.032	0.000	0.000	50
RCD	0.541	0.547	0.807	0.820	50

Table 16: Pair-subsystem comparisons whose sign, or whose reversal status, changes between the fixed release and the release as shipped. Δ fixed is the reference (header repaired); Δ as shipped is the alternative.

Layer	Pair	Subsystem	Δ fixed [CI]	Δ as shipped [CI]
Published	CausalRCA vs CIRCA	RE1-OB	-0.056 [-0.144, +0.035]	+0.184 [+0.077, +0.291]
Published	CIRCA vs RCD	RE1-OB	+0.131 [+0.021, +0.237]	-0.115 [-0.245, +0.016]
All	CausalRCA vs CIRCA	RE1-OB	-0.056 [-0.144, +0.035]	+0.184 [+0.077, +0.291]
All	CIRCA vs max- Z	RE1-OB	+0.168 [+0.064, +0.280]	-0.072 [-0.192, +0.048]
All	CIRCA vs max- Z	RE1-SS	-0.040 [-0.144, +0.064]	-0.040 [-0.144, +0.064]
All	CIRCA vs max- Z	RE1-TT	-0.096 [-0.192, +0.008]	-0.096 [-0.192, +0.008]
All	CIRCA vs RCD	RE1-OB	+0.131 [+0.021, +0.237]	-0.115 [-0.245, +0.016]

Published tables

Pooled margin against between-stratum dispersion for every pair inside two published tables. Source: `paper/data/rq1/published_tables_margin_dispersion.csv`.

Table 17: Margin and between-stratum dispersion inside two published tables (RCAEval Table 6 and PetShop Table 8). Below line marks $|\text{pooled margin}| < \text{SD}$ across strata. #strata reversing is the number of strata whose sign disagrees with the pooled margin.

Table	Pair	k	Pooled	margin	SD	Below line	#strata reversing
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	BARO (metric) vs CausalRCA (metric)	6	+0.447	0.447	0.255		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	BARO (metric) vs CIRCA (metric)	6	+0.343	0.343	0.106		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	BARO (metric) vs MicroCause (metric)	6	+0.568	0.568	0.213		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	BARO (metric) vs RCD (metric)	6	+0.578	0.578	0.282		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	BARO (metric) vs MicroRank (trace)	6	+0.503	0.503	0.284		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	BARO (metric) vs TraceRCA (trace)	6	+0.012	0.012	0.281	✓	3
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	BARO (metric) vs BARO (multi)	6	-0.023	0.023	0.057	✓	0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	BARO (metric) vs CIRCA (multi)	6	+0.610	0.610	0.256		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	BARO (metric) vs PDiagnose (multi)	6	+0.190	0.190	0.401	✓	2
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	BARO (metric) vs RCD (multi)	6	+0.572	0.572	0.283		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	CausalRCA (metric) vs CIRCA (metric)	6	-0.103	0.103	0.164	✓	1
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	CausalRCA (metric) vs MicroCause (metric)	6	+0.122	0.122	0.086		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	CausalRCA (metric) vs RCD (metric)	6	+0.132	0.132	0.182	✓	1
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	CausalRCA (metric) vs MicroRank (trace)	6	+0.057	0.057	0.221	✓	3
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	CausalRCA (metric) vs TraceRCA (trace)	6	-0.435	0.435	0.197		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	CausalRCA (metric) vs BARO (multi)	6	-0.470	0.470	0.267		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	CausalRCA (metric) vs CIRCA (multi)	6	+0.163	0.163	0.212	✓	1
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	CausalRCA (metric) vs PDiagnose (multi)	6	-0.257	0.257	0.280	✓	1
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	CausalRCA (metric) vs RCD (multi)	6	+0.125	0.125	0.146	✓	1
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	CIRCA (metric) vs MicroCause (metric)	6	+0.225	0.225	0.137		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	CIRCA (metric) vs RCD (metric)	6	+0.235	0.235	0.185		0

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Table 17: Margin and between-stratum dispersion inside two published tables (RCAEval Table 6 and PetShop Table 8). Below line marks |pooled margin| < SD across strata. #strata reversing is the number of strata whose sign disagrees with the pooled margin.

Table	Pair	k	Pooled	margin	SD	Below line	#strata reversing
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	CIRCA (metric) vs MicroRank (trace)	6	+0.160	0.160	0.199	✓	1
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	CIRCA (metric) vs TraceRCA (trace)	6	-0.332	0.332	0.191		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	CIRCA (metric) vs BARO (multi)	6	-0.367	0.367	0.117		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	CIRCA (metric) vs CIRCA (multi)	6	+0.267	0.267	0.173		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	CIRCA (metric) vs PDiagnose (multi)	6	-0.153	0.153	0.311	✓	2
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	CIRCA (metric) vs RCD (multi)	6	+0.228	0.228	0.186		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	MicroCause (metric) vs RCD (metric)	6	+0.010	0.010	0.203	✓	4
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	MicroCause (metric) vs MicroRank (trace)	6	-0.065	0.065	0.250	✓	1
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	MicroCause (metric) vs TraceRCA (trace)	6	-0.557	0.557	0.203		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	MicroCause (metric) vs BARO (multi)	6	-0.592	0.592	0.216		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	MicroCause (metric) vs CIRCA (multi)	6	+0.042	0.042	0.208	✓	4
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	MicroCause (metric) vs PDiagnose (multi)	6	-0.378	0.378	0.300		1
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	MicroCause (metric) vs RCD (multi)	6	+0.003	0.003	0.179	✓	3
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	RCD (metric) vs MicroRank (trace)	6	-0.075	0.075	0.091	✓	1
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	RCD (metric) vs TraceRCA (trace)	6	-0.567	0.567	0.096		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	RCD (metric) vs BARO (multi)	6	-0.602	0.602	0.276		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	RCD (metric) vs CIRCA (multi)	6	+0.032	0.032	0.066	✓	1
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	RCD (metric) vs PDiagnose (multi)	6	-0.388	0.388	0.194		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	RCD (metric) vs RCD (multi)	6	-0.007	0.007	0.069	✓	2
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	MicroRank (trace) vs TraceRCA (trace)	6	-0.492	0.492	0.177		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	MicroRank (trace) vs BARO (multi)	6	-0.527	0.527	0.279		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	MicroRank (trace) vs CIRCA (multi)	6	+0.107	0.107	0.111	✓	0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	MicroRank (trace) vs PDiagnose (multi)	6	-0.313	0.313	0.267		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	MicroRank (trace) vs RCD (multi)	6	+0.068	0.068	0.105	✓	1
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	TraceRCA (trace) vs BARO (multi)	6	-0.035	0.035	0.281	✓	2
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	TraceRCA (trace) vs CIRCA (multi)	6	+0.598	0.598	0.116		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	TraceRCA (trace) vs PDiagnose (multi)	6	+0.178	0.178	0.135		1
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	TraceRCA (trace) vs RCD (multi)	6	+0.560	0.560	0.144		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	BARO (multi) vs CIRCA (multi)	6	+0.633	0.633	0.239		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	BARO (multi) vs PDiagnose (multi)	6	+0.213	0.213	0.405	✓	2
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	BARO (multi) vs RCD (multi)	6	+0.595	0.595	0.277		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	CIRCA (multi) vs PDiagnose (multi)	6	-0.420	0.420	0.231		0
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	CIRCA (multi) vs RCD (multi)	6	-0.038	0.038	0.103	✓	3
RCAEval Table 6 (RE2-TT, AC@1, 6 fault types)	PDiagnose (multi) vs RCD (multi)	6	+0.382	0.382	0.242		0
PetShop Table 8 (top-1, 6 strata)	Traversal vs CIRCA	6	+0.157	0.157	0.126		0
PetShop Table 8 (top-1, 6 strata)	Traversal vs Counterfactual	6	+0.340	0.340	0.218		0
PetShop Table 8 (top-1, 6 strata)	Traversal vs ϵ -Diagnosis	6	+0.538	0.538	0.194		0
PetShop Table 8 (top-1, 6 strata)	Traversal vs RCD	6	+0.325	0.325	0.356	✓	2
PetShop Table 8 (top-1, 6 strata)	Traversal vs Ranked correlation	6	-0.090	0.090	0.313	✓	2
PetShop Table 8 (top-1, 6 strata)	CIRCA vs Counterfactual	6	+0.183	0.183	0.228	✓	1
PetShop Table 8 (top-1, 6 strata)	CIRCA vs ϵ -Diagnosis	6	+0.382	0.382	0.211		0
PetShop Table 8 (top-1, 6 strata)	CIRCA vs RCD	6	+0.168	0.168	0.301	✓	2
PetShop Table 8 (top-1, 6 strata)	CIRCA vs Ranked correlation	6	-0.247	0.247	0.326	✓	1
PetShop Table 8 (top-1, 6 strata)	Counterfactual vs ϵ -Diagnosis	6	+0.198	0.198	0.239	✓	0
PetShop Table 8 (top-1, 6 strata)	Counterfactual vs RCD	6	-0.015	0.015	0.450	✓	3
PetShop Table 8 (top-1, 6 strata)	Counterfactual vs Ranked correlation	6	-0.430	0.430	0.345		0
PetShop Table 8 (top-1, 6 strata)	ϵ -Diagnosis vs RCD	6	-0.213	0.213	0.259	✓	0
PetShop Table 8 (top-1, 6 strata)	ϵ -Diagnosis vs Ranked correlation	6	-0.628	0.628	0.149		0
PetShop Table 8 (top-1, 6 strata)	RCD vs Ranked correlation	6	-0.415	0.415	0.274		0

Fault composition

Case counts by fault class, and the comparisons whose sign changes after re-weighting fault classes. Source: [paper/data/rq2/fault_composition.csv](#).

Table 18: Case counts per fault class and subsystem. Fault classes that are zero in every subsystem are omitted.

Subsystem	availability	cpu	delay	disk	latency	loss	mem	memory	network	other	socket
RE1-OB	0	25	25	25	0	25	25	0	0	0	0
RE1-SS	0	25	25	25	0	25	25	0	0	0	0
RE1-TT	0	25	25	25	0	25	25	0	0	0	0
RE2-OB	0	15	15	15	0	15	15	0	0	0	15
RE2-SS	0	15	15	15	0	15	15	0	0	0	15
RE2-TT	0	15	15	15	0	15	15	0	0	0	15
Bank	0	36	0	24	0	0	0	17	59	0	0
Market-1	0	13	0	24	0	0	0	12	17	4	0
Market-2	0	13	0	27	0	0	0	10	25	3	0
Telecom	0	19	0	0	0	0	0	0	20	12	0
High	12	0	0	0	14	0	0	0	0	0	0
Low	12	0	0	0	14	0	0	0	0	0	0
Temporal-1	4	0	0	0	4	0	0	0	0	0	0

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Table 18: Case counts per fault class and subsystem. Fault classes that are zero in every subsystem are omitted.

Subsystem	availability	cpu	delay	disk	latency	loss	mem	memory	network	other	socket
Temporal-2	4	0	0	0	4	0	0	0	0	0	0

Table 19: Comparisons whose sign changes when each fault class is re-weighted to a common mix. Δ is the original per-subsystem difference; Δ re-weighted is the re-weighted difference with its 95% CI.

Family	Layer	Pair	Subsystem	Δ	Δ re-weighted [CI]
OpenRCA	Published	BARO vs ϵ -Diagnosis	Market-1	+0.019	-0.006 [-0.067, +0.054]
OpenRCA	Published	ϵ -Diagnosis vs RCD	Market-1	-0.013	+0.017 [-0.052, +0.081]
OpenRCA	Published	ϵ -Diagnosis vs RCD	Market-2	-0.023	+0.039 [-0.033, +0.116]
OpenRCA	All	Alert-count vs BARO	Telecom	+0.059	-0.004 [-0.037, +0.027]
OpenRCA	All	Alert-count vs CD-1min	Bank	+0.007	-0.030 [-0.121, +0.061]
OpenRCA	All	Alert-count vs CD-1min	Market-1	-0.017	+0.052 [-0.025, +0.144]
OpenRCA	All	Alert-count vs CD-1min	Market-2	+0.004	-0.010 [-0.146, +0.116]
OpenRCA	All	Alert-count vs max- Z	Telecom	+0.010	-0.001 [-0.090, +0.094]
OpenRCA	All	Alert-count vs RCD	Telecom	+0.049	-0.010 [-0.066, +0.040]
OpenRCA	All	BARO vs ϵ -Diagnosis	Market-1	+0.019	-0.006 [-0.067, +0.054]
OpenRCA	All	BARO vs max- Z	Telecom	-0.049	+0.003 [-0.086, +0.094]
OpenRCA	All	CD-1min vs max- Z	Market-1	+0.000	-0.088 [-0.214, +0.030]
OpenRCA	All	CD-1min vs max- Z	Market-2	-0.007	+0.000 [-0.145, +0.160]
OpenRCA	All	ϵ -Diagnosis vs RCD	Market-1	-0.013	+0.017 [-0.052, +0.081]
OpenRCA	All	ϵ -Diagnosis vs RCD	Market-2	-0.023	+0.039 [-0.033, +0.116]
OpenRCA	All	max- Z vs RCD	Telecom	+0.039	-0.009 [-0.099, +0.082]
PetShop	Published	BARO vs Counterfactual	Low	+0.000	+0.018 [-0.173, +0.208]
PetShop	All	BARO vs Counterfactual	Low	+0.000	+0.018 [-0.173, +0.208]
PetShop	All	Traversal vs Ranked correlation	Low	+0.000	-0.012 [-0.238, +0.208]

Surveyed papers

The paper list with the per-system reporting check and the three papers excluded for lack of full text. Source: `paper/data/survey/papers.csv`.

Table 20: The surveyed papers. Per-system results: Yes if the paper evaluates more than one system and reports a score for each; Single system if it evaluates one system; Excluded if no full text was obtained (not counted in the paper). Location: the table, figure, or passage checked. Venue names longer than 30 characters are truncated. The italic line under each row is the paper title.

ID	Year	Venue	Per-system results	Location
baro2024	2024	FSE 2024 (Proc. ACM Softw...	Yes	Table 2 caption (cf. Table 1/3/4 captions, all "three datasets"/"three benchmark microservice systems")
<i>BARO: Robust Root Cause Analysis for Microservices via Multivariate Bayesian Online Change Point Detection</i>				
circa2022	2022	KDD 2022	Yes	Table 1 caption; cf. Table 3 caption "Performance of different methods on D_O"
<i>Causal Inference-Based Root Cause Analysis for Online Service Systems with Intervention Recognition</i>				
rcd2022	2022	NeurIPS 2022	Yes	Table 1 caption; cf. Table 2 caption "summary and top-7 recall of epsilon-Diagnosis and RCD on data collected for three outages from a..."
<i>Root Cause Analysis of Failures in Microservices through Causal Discovery</i>				
tracerca2021	2021	IWQoS 2021	Yes	Sec. IV-B (discussion of Table II)
<i>Practical Root Cause Localization for Microservice Systems via Trace Analysis</i>				
microrank2021	2021	WWW 2021	Excluded (no full text)	–
<i>MicroRank: End-to-End Latency Issue Localization with Extended Spectrum Analysis in Microservice Environments</i>				
raeval2025	2025	WWW 2025 (Companion...	Single system	Table 6 caption, Sec 5 (p.4)
<i>RCAEval: A Benchmark for Root Cause Analysis of Microservice Systems with Telemetry Data</i>				
fang2026	2026	FSE 2026 (Proc. ACM Softw...	Yes	Sec. 3.2.2 Answering RQ2
<i>Rethinking the Evaluation of Microservice RCA with a Fault Propagation-Aware Benchmark</i>				
openrca2025	2025	ICLR 2025	Yes	Text at the discussion of Figure 5
<i>OpenRCA: Can Large Language Models Locate the Root Cause of Software Failures?</i>				
petshop2024	2024	CLear 2024	Yes	Table 3
<i>The PetShop Dataset — Finding Causes of Performance Issues across Microservices</i>				
eadro2023	2023	ICSE 2023	Yes	Sec. V-E RQ1 (Table II lists TT and SN side by side)
<i>Eadro: An End-to-End Troubleshooting Framework for Microservices on Multi-source Data</i>				
nezha2023	2023	ESEC/FSE 2023	Excluded (no full text)	–
<i>Nezha: Interpretable Fine-Grained Root Causes Analysis for Microservices on Multi-modal Observability Data</i>				
mabc2024	2024	EMNLP 2024 (Findings)	Yes	Table 4
<i>mABC: Multi-Agent Blockchain-inspired Collaboration for Root Cause Analysis in Micro-Services Architecture</i>				
mulan2024	2024	WWW 2024	Yes	Table 2 caption; cf. Table 3 (Online Boutique)/Table 4 (Train Ticket) captions
<i>MULAN: Multi-modal Causal Structure Learning and Root Cause Analysis</i>				
ocean2024	2025	IEEE BigData 2025	Yes	Table II caption (Table III: Online Boutique; Table V: Train Ticket, same format)

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Table 20: The surveyed papers. Per-system results: Yes if the paper evaluates more than one system and reports a score for each; Single system if it evaluates one system; Excluded if no full text was obtained (not counted in the paper). Location: the table, figure, or passage checked. Venue names longer than 30 characters are truncated. The italic line under each row is the paper title.

ID	Year	Venue	Per-system results	Location
<i>Online Multi-modal Root Cause Identification in Microservice Systems (OCEAN)</i>				
tvdiag2024	2025	ACM TOSEM	Yes	Discussion of Table 4
<i>TVDiag: A Task-oriented and View-invariant Failure Diagnosis Framework for Microservice-based Systems with Multimodal Data</i>				
mrca2024	2024	ASE 2024	Excluded (no full text)	–
<i>MRCA: Metric-level Root Cause Analysis for Microservices via Multi-Modal Data</i>				
chase2024	2024	arXiv preprint	Yes	Text at the discussion of Table II (overall performance comparison, one row per dataset)
<i>CHASE: A Causal Hypergraph based Framework for Root Cause Analysis in Multimodal Microservice Systems</i>				
howfar2024	2024	ASE 2024	Yes	Table 5 caption
<i>Root Cause Analysis for Microservice System based on Causal Inference: How Far Are We?</i>				
lemmarca2024	2024	OpenReview	Yes	Table 4 caption (cf. Table 3: multi-modality Product Review/Cloud Computing results; Table 5: WADI; Table 6: caption line 793)
<i>LEMMA-RCA: A Large Multi-modal Multi-domain Dataset for Root Cause Analysis</i>				
sparserca2024	2024	ISSRE 2024	Single system	Sec. VI.A.1 Datasets, p.9
<i>SparseRCA: Unsupervised Root Cause Analysis in Sparse Microservice Testing Traces</i>				
tracecontrast2024	2024	ICSE 2024	Single system	Sec. 4.1.1 Dataset, p.7
<i>Trace-based Multi-Dimensional Root Cause Localization of Performance Issues in Microservice Systems</i>				
tracediag2023	2023	ESEC/FSE 2023	Single system	Sec. 5.1.1 Dataset, p.13
<i>TraceDiag: Adaptive, Interpretable, and Efficient Root Cause Analysis on Large-Scale Microservice Systems</i>				
run2024	2024	AAAI 2024	Yes	Text (Tables 2 and 3: sock-shop; Figure 6: synthetic dataset, reported separately)
<i>Root Cause Analysis In Microservice Using Neural Granger Causal Discovery</i>				
dynacausal2025	2025	arXiv preprint	Yes	Table 2 caption (datasets D1 and D2)
<i>DynaCausal: Dynamic Causality-Aware Root Cause Analysis for Distributed Microservices</i>				
causalrca2023	2023	Journal of Systems and...	Single system	Sec. I Introduction, p.2 (also Sec. Threats to Validity)
<i>CausalRCA: Causal inference based precise fine-grained root cause localization for microservice applications</i>				
toomanycooks2025	2025	ISSRE 2025	Yes	Table IV caption
<i>Too Many Cooks: Assessing the Need for Multi-Source Data in Microservice Failure Diagnosis</i>				

Cross-release, all pairs

RE1 versus RE2 differences for every pair, including probes and shipped baselines. Source: [paper/data/rq1/cross_release.csv](#).

Source:

Table 21: Cross-release comparison for every pair, including probes (the all-pairs layer). Δ is accuracy on the RE1 release minus the paired method, and the same contrast on the RE2 release, each with its 95% CI. Same sign marks agreement of the two differences; CIs opposite marks intervals that lie on opposite sides of zero.

Pair	App	Δ RE1 [CI]	Δ RE2 [CI]	Same sign	CIs opposite
Alert-count vs BARO	OB	−0.344 [−0.440, −0.248]	+0.067 [−0.056, +0.189]		
	SS	−0.472 [−0.576, −0.360]	+0.322 [+0.211, +0.433]		✓
	TT	−0.512 [−0.608, −0.408]	−0.456 [−0.567, −0.344]	✓	
Alert-count vs CausalRCA	OB	−0.224 [−0.320, −0.128]	+0.004 [−0.111, +0.122]		
	SS	+0.200 [+0.099, +0.301]	+0.241 [+0.130, +0.356]	✓	
	TT	−0.088 [−0.144, −0.029]	+0.004 [−0.089, +0.100]		
Alert-count vs CD-1min	OB	−0.336 [−0.448, −0.224]	−0.522 [−0.644, −0.389]	✓	
	SS	−0.552 [−0.648, −0.448]	−0.556 [−0.656, −0.456]	✓	
	TT	−0.232 [−0.328, −0.144]	−0.578 [−0.689, −0.467]	✓	
Alert-count vs CIRCA	OB	−0.280 [−0.392, −0.176]	−0.456 [−0.589, −0.322]	✓	
	SS	−0.328 [−0.448, −0.208]	−0.389 [−0.511, −0.267]	✓	
	TT	−0.280 [−0.368, −0.192]	−0.333 [−0.456, −0.211]	✓	
Alert-count vs ϵ -Diagnosis	OB	+0.360 [+0.272, +0.448]	+0.178 [+0.089, +0.267]	✓	
	SS	+0.232 [+0.120, +0.344]	+0.233 [+0.111, +0.356]	✓	
	TT	+0.048 [+0.016, +0.088]	+0.211 [+0.133, +0.300]	✓	
Alert-count vs max- Z	OB	−0.112 [−0.224, +0.000]	−0.567 [−0.689, −0.433]	✓	
	SS	−0.368 [−0.488, −0.248]	−0.456 [−0.589, −0.322]	✓	
	TT	−0.376 [−0.472, −0.280]	−0.200 [−0.322, −0.067]	✓	
BARO vs CausalRCA	OB	+0.120 [+0.043, +0.195]	−0.063 [−0.159, +0.041]		
	SS	+0.672 [+0.592, +0.752]	−0.081 [−0.163, −0.004]		✓
	TT	+0.424 [+0.331, +0.515]	+0.459 [+0.341, +0.574]	✓	
BARO vs CD-1min	OB	+0.008 [−0.080, +0.096]	−0.589 [−0.700, −0.478]		
	SS	−0.080 [−0.136, −0.024]	−0.878 [−0.944, −0.811]	✓	
	TT	+0.280 [+0.160, +0.392]	−0.122 [−0.233, −0.011]		✓
BARO vs CIRCA	OB	+0.064 [−0.032, +0.160]	−0.522 [−0.644, −0.400]		
	SS	+0.144 [+0.064, +0.232]	−0.711 [−0.800, −0.611]		✓
	TT	+0.232 [+0.136, +0.328]	+0.122 [+0.011, +0.233]	✓	
BARO vs ϵ -Diagnosis	OB	+0.704 [+0.624, +0.784]	+0.111 [+0.022, +0.200]	✓	
	SS	+0.704 [+0.616, +0.784]	−0.089 [−0.189, +0.000]		
	TT	+0.560 [+0.472, +0.640]	+0.667 [+0.567, +0.767]	✓	
BARO vs max- Z	OB	+0.232 [+0.136, +0.328]	−0.633 [−0.733, −0.522]		✓

Continued on next page

Table 21: Cross-release comparison for every pair, including probes (the all-pairs layer). Δ is accuracy on the RE1 release minus the paired method, and the same contrast on the RE2 release, each with its 95% CI. Same sign marks agreement of the two differences; CIs opposite marks intervals that lie on opposite sides of zero.

Pair	App	Δ RE1 [CI]	Δ RE2 [CI]	Same sign	CIs opposite
CausalRCA vs CD-1min	SS	+0.104 [+0.032, +0.176]	-0.778 [-0.856, -0.689]		✓
	TT	+0.136 [+0.048, +0.224]	+0.256 [+0.133, +0.378]	✓	
	OB	-0.112 [-0.197, -0.027]	-0.526 [-0.619, -0.430]	✓	
	SS	-0.752 [-0.819, -0.680]	-0.796 [-0.867, -0.722]	✓	
CausalRCA vs CIRCA	TT	-0.144 [-0.235, -0.056]	-0.581 [-0.678, -0.481]	✓	
	OB	-0.056 [-0.144, +0.035]	-0.459 [-0.567, -0.356]	✓	
	SS	-0.528 [-0.613, -0.440]	-0.630 [-0.722, -0.530]	✓	
	TT	-0.192 [-0.280, -0.104]	-0.337 [-0.456, -0.219]	✓	
CausalRCA vs ϵ -Diagnosis	OB	+0.584 [+0.512, +0.659]	+0.174 [+0.089, +0.259]	✓	
	SS	+0.032 [-0.051, +0.117]	-0.007 [-0.104, +0.089]		
	TT	+0.136 [+0.096, +0.179]	+0.207 [+0.152, +0.267]	✓	
	OB	+0.112 [+0.008, +0.216]	-0.570 [-0.667, -0.470]		✓
CausalRCA vs max- Z	SS	-0.568 [-0.661, -0.475]	-0.696 [-0.789, -0.596]	✓	
	TT	-0.288 [-0.379, -0.195]	-0.204 [-0.311, -0.100]	✓	
	OB	+0.056 [-0.040, +0.152]	+0.067 [-0.044, +0.178]	✓	
	SS	+0.224 [+0.144, +0.304]	+0.167 [+0.089, +0.256]	✓	
CD-1min vs CIRCA	TT	-0.048 [-0.160, +0.072]	+0.244 [+0.133, +0.356]		
	OB	+0.696 [+0.616, +0.776]	+0.700 [+0.600, +0.800]	✓	
	SS	+0.784 [+0.712, +0.856]	+0.789 [+0.689, +0.878]	✓	
	TT	+0.280 [+0.200, +0.360]	+0.789 [+0.700, +0.867]	✓	
CD-1min vs ϵ -Diagnosis	OB	+0.224 [+0.120, +0.336]	-0.044 [-0.122, +0.033]		
	SS	+0.184 [+0.112, +0.256]	+0.100 [+0.022, +0.178]	✓	
	TT	-0.144 [-0.248, -0.032]	+0.378 [+0.256, +0.500]		✓
	OB	+0.640 [+0.552, +0.720]	+0.633 [+0.533, +0.744]	✓	
CIRCA vs ϵ -Diagnosis	SS	+0.560 [+0.464, +0.648]	+0.622 [+0.500, +0.733]	✓	
	TT	+0.328 [+0.248, +0.408]	+0.544 [+0.444, +0.644]	✓	
	OB	+0.168 [+0.064, +0.280]	-0.111 [-0.211, -0.011]		✓
	SS	-0.040 [-0.144, +0.064]	-0.067 [-0.156, +0.011]	✓	
CIRCA vs max- Z	TT	-0.096 [-0.192, +0.008]	+0.133 [+0.000, +0.267]		
	OB	-0.472 [-0.560, -0.376]	-0.744 [-0.833, -0.656]	✓	
	SS	-0.600 [-0.696, -0.496]	-0.689 [-0.800, -0.567]	✓	
	TT	-0.424 [-0.512, -0.336]	-0.411 [-0.511, -0.311]	✓	
ϵ -Diagnosis vs max- Z	OB	-0.472 [-0.560, -0.376]	-0.744 [-0.833, -0.656]	✓	
	SS	-0.600 [-0.696, -0.496]	-0.689 [-0.800, -0.567]	✓	
	TT	-0.424 [-0.512, -0.336]	-0.411 [-0.511, -0.311]	✓	
	OB	-0.472 [-0.560, -0.376]	-0.744 [-0.833, -0.656]	✓	

Table 5: Cross-family set, held-out selection: for each of the 11 subsystems, the pooled winner of the other ten (subsystems weighted equally) is applied to it. Regret: accuracy lost on the held-out subsystem.

Pair	Reversals	Mean regret	Max regret	Held-out subsystems where the pooled winner loses
alert-count vs BARO	4 of 11	0.023	0.091	Bank, Market-1, Market-2, Telecom
alert-count vs CD-1min	5 of 11	0.023	0.125	Bank, Market-2, Telecom, High, Temporal-2
alert-count vs ϵ -Diagnosis	0 of 11	0.000	0.000	
alert-count vs max- Z	2 of 11	0.005	0.045	Bank, Telecom
alert-count vs RCD	5 of 11	0.036	0.130	Bank, Market-1, Market-2, Telecom, High
BARO vs CD-1min	4 of 11	0.025	0.104	RE1-SS, Bank, Market-1, Market-2
BARO vs ϵ -Diagnosis	0 of 11	0.000	0.000	
BARO vs max- Z	10 of 11	0.099	0.232	RE1-OB, RE1-SS, RE1-TT, Market-1, Market-2, Telecom, High, Low, Temporal-1, Temporal-2
BARO vs RCD	4 of 11	0.064	0.375	Telecom, Low, Temporal-1, Temporal-2
CD-1min vs ϵ -Diagnosis	0 of 11	0.000	0.000	
CD-1min vs max- Z	4 of 11	0.041	0.224	RE1-OB, RE1-SS, Bank, Market-1
CD-1min vs RCD	7 of 11	0.182	0.539	RE1-OB, RE1-SS, RE1-TT, Telecom, Low, Temporal-1, Temporal-2
ϵ -Diagnosis vs max- Z	0 of 11	0.000	0.000	
ϵ -Diagnosis vs RCD	0 of 11	0.000	0.000	
max- Z vs RCD	3 of 11	0.040	0.250	RE1-OB, Low, Temporal-1

Table 6: Cross-family set, robustness. Drop one family: subsystems where A / B is more accurate among the remaining ones. Drop two: cases (out of all 55 ways to drop two subsystems) in which both signs remain. Pooled difference under four pooling rules (trimmed: the largest and smallest subsystem differences removed). Bootstrap: share of 2,000 case resamples in which both signs remain.

Pair	Drop one family	Drop two	Mean	Median	Case-wtd.	Trimmed	Bootstrap
alert-count vs BARO	RCAEval RE1: 4/3; OpenRCA: 0/6; PetShop: 4/3	55/55	-0.145	-0.115	-0.202	-0.131	1.000
alert-count vs CD-1min	RCAEval RE1: 5/3; OpenRCA: 2/5; PetShop: 3/4	55/55	-0.095	-0.017	-0.175	-0.069	0.999
alert-count vs ϵ -Diagnosis	RCAEval RE1: 6/0; OpenRCA: 5/0; PetShop: 7/0	0/55	+0.127	+0.125	+0.167	+0.115	0.185
alert-count vs max- Z	RCAEval RE1: 2/6; OpenRCA: 0/7; PetShop: 2/5	54/55	-0.141	-0.112	-0.144	-0.136	0.997
alert-count vs RCD	RCAEval RE1: 5/3; OpenRCA: 1/6; PetShop: 4/3	55/55	-0.082	-0.013	-0.011	-0.046	1.000
BARO vs CD-1min	RCAEval RE1: 5/3; OpenRCA: 6/1; PetShop: 3/4	55/55	+0.050	+0.020	+0.027	+0.042	1.000
BARO vs ϵ -Diagnosis	RCAEval RE1: 8/0; OpenRCA: 7/0; PetShop: 7/0	0/55	+0.272	+0.154	+0.369	+0.252	0.340
BARO vs max- Z	RCAEval RE1: 2/6; OpenRCA: 4/3; PetShop: 4/3	55/55	+0.004	-0.038	+0.058	-0.007	1.000
BARO vs RCD	RCAEval RE1: 4/4; OpenRCA: 4/3; PetShop: 6/1	55/55	+0.063	+0.040	+0.191	+0.066	1.000
CD-1min vs ϵ -Diagnosis	RCAEval RE1: 6/0; OpenRCA: 5/0; PetShop: 7/0	0/55	+0.222	+0.125	+0.342	+0.184	0.361
CD-1min vs max- Z	RCAEval RE1: 2/6; OpenRCA: 2/5; PetShop: 4/3	55/55	-0.046	-0.069	+0.031	-0.053	1.000
CD-1min vs RCD	RCAEval RE1: 3/4; OpenRCA: 3/3; PetShop: 6/1	55/55	+0.013	+0.076	+0.163	+0.011	1.000
ϵ -Diagnosis vs max- Z	RCAEval RE1: 0/8; OpenRCA: 0/7; PetShop: 0/7	0/55	-0.268	-0.192	-0.311	-0.252	0.086
ϵ -Diagnosis vs RCD	RCAEval RE1: 0/7; OpenRCA: 0/6; PetShop: 0/7	0/55	-0.209	-0.147	-0.178	-0.186	0.469
max- Z vs RCD	RCAEval RE1: 5/2; OpenRCA: 3/3; PetShop: 6/1	55/55	+0.059	+0.039	+0.132	+0.060	0.997